Trenitalia
ETR1000 VZI-6

High Voltage

HITACHI Inspire the Next	**Created:**	**21/07/2025**	**P. Buda**
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Revision	Author	Date	Revision Description
00.1	R. Schiavo	03/03/2023	Draft Release Platform-Zefiro-Europe /2F_04.01.-.Line Voltage-VF.-.Zefiro-Europe/Line Voltage-VF.CONCEPT, current 9.0, doc 3EGH489028-2358_B reviewed for VZI-6
00	R. Schiavo		First Emission
01.1			Added detailed requiremnts for line Voltage supervision: 2F_04.01.Zefiro-Europe.CONCEPT.HR06-2F_04.01.Zefiro-Europe.CONCEPT.HR15 line Current supervision: 2F_04.01.Zefiro-Europe.CONCEPT.HR16-2F_04.01.Zefiro-Europe.CONCEPT.HR17 High Voltage system Protection: self test requirements review Modified energy meter requirements due to architecture change (JC-K001 eliminated) Added “Main Transformer cooling and supervision” requirements
1.0	R. Schiavo	21/09/2023	Release Emission
2.1	R. Schiavo	07/11/2023	Updating after TCMS SW designer review Modified: .HR04,.1329,.786,.216,.HR11,.HR14,.HR15,.989,.HR29,.1319,.755,.412,.1082,.1087,.1088,,.HR20,.1069,.HR38,,.HR39,.414,.HR29,.1054,.1055 New Requirements: .HR170,.HR171,.HR172,.HR173,.HR174
2.2	R. Schiavo	09/11/2023	Modified: .989, . New Requirements: .HR175,.HR176,.HR177,.HR178,.HR179,.HR180,

2.3	R. Schiavo	27/03/2024	Modified (15kV): .670,.1108,.786,.1050,.1152,.374 Added (15kV): TRS.705,.350 Modified: HR03,.1160(asVZES),.HR12,.HR13,.HR71,.HR177,.HR150,.HR18, .HR170, .758,.106,.HR17,.759,.209,.234 .989,HR173,.HR178,.HR179,.HR180,.HR177,.HR176,.(HVSP self test) New Requirements: HR181,HR182,.HR183,.HR184,(Differential current protection) .HR185, .HR186,.HR187,.HR192, .HR193, .HR194,.1032 ,1036, HR195,.HR196,HR197,HR198,HR199,HR200,HR201,HR203,HR204, .HR186 (HVSP line trip opening request) .HR212,.HR213,.HR214,.HR218,.HR219,.(HVSP Diagnostic) HR210,.HR211 (Undervoltage level1) .HR205,.HR206 (Line current monitoring) .HR207,.HR208 (AC line over current) .HR226 (pantograph up status condition) .HR227-.HR230 (Transforme temperature sensor failure) .HR243,.HR241,.HR232,.HR234,.HR236,.HR242,.HR231,.HR175,.HR202, HR239,.HR237,.HR235 (HVSP self Test) .HR244-HR252 (added to LCB closing) .HR253--.HR256 (added to High Voltage Isolating, opening (closing roof disconnectors timing) .HR257--.HR259, .HR263 (added to line current supervision, LCB opening) .HR260-.HR262 (AC VMT failure) .HR264 (50Hz bypass) .HR265 (Neutral Section) .HR188,.HR189,.HR190, .HR191,.HR266 (2° LCB closing) Deleted HR05,.314
2.4	R. Schiavo	28/03/2024	Review HVSP functional requirements with HVSP designer
2.5	R. Schiavo	29/03/2024	Modified Requirements: .HR13,.HR2024,.HR09,.HR214,.313,.315,.319,.316,503,.317,.HR210,.318,. 541 (Line Voltage Supervision) .1129,.225 (Line Current Supervision) .234 (High Voltage System Protection) .HR29 (LCB Closing) .1105 (LCB Opening) .1000 (Isolate HV) New Requirements: .HR281,.HR267,.HR269,.HR270,.FR271,.HR268,.HR272,.HR274,.HR275 (Line Voltage Supervision) .HR276,.HR277,. (Line Current Supervision) .HR 285,.HR286,.HR288 (High Voltage System Protection) .HR284 (Neutral Section) .HR280,.HR282 (LCB Closing) .HR283,.HR289 (Main Transformer Supervision) Deleted .362,.464 Added Functional requirement references to Interface signals: HVSP->CCU, CCU->HVSP
2.6	R. Schiavo	02/04/2024	Added Functional requirement references to all Interface signals
2.7	R. Schiavo	09/04/2024	New Requirements: .HR310, .HR311,.HR312,.HR313,.

2.8	R. Schiavo	12/04/2024	Eliminata modifica sul valore in tensione di setting del contatore 2F_04.01.Zefiro-Europe.CONCEPT.503
2.9	R. Schiavo	23/04/2024	Modified 2F_04.01.Zefiro-Europe.CONCEPT.1195 Added 2F_04.01.Zefiro-Europe.CONCEPT.HR314,.HR315,.HR316, .HR317 (filter precharge sequence)
2.0	R. Schiavo	29/04/2024	Emission - BL1 baseline
3.1	R. Schiavo	30/07/2024	Modified Requiremnts: .1216, .361,.HR16,.HR17,.HR257,.776,.526,.527,.785,.1000,.HR62,.HR228, .497,.498,.422,.755 Deleted Requirements: .HR258,.HR259,.HR230,.HR216 New Requirements: .HR322,..HR320,.HR321,.HR324,.HR323,.HR327,.HR325,.HR326
3.2	R. Schiavo	02/08/2024	Allocation change req .762 form TCMS to PCU Modified requirement HR227 (transformer pt!00 range)
3.3	R. Schiavo	12/09/2024	HVSP diagnostic: Eliminated requirements .HR270,.HR213 Modified requirements: .HR272,.HR214,.HR274,.HR218 Panto up to HVSP and Propulsion not set if the panto is udsed for ice scrapig: Modified requirements: .749,.HR04
3.0	R. Schiavo	16/09/2024	Emission – BL2 baseline
4.1	R. Schiavo	10/10/2024	Modified requirements .106, .HR288,.HR284 Added requirement .HR328
4.2	R. Schiavo	14/10/2024	Modifies Req .361, .HR277 Added code of the requirements reported in the .HR328 requirement condition Req .HR276,.HR277 Added that the current threshold is evaluated as Arms
4.3	R. Schiavo	15/10/2024	Deleted Requirements: .HR13,.HR204 Modified Requirements: .HR280,.HR282,.(Master/slave Condition in DC) HR210,.583,.541,.225,.315,.316,.317,.318 (protection activated in Master status) HR226(panto up status in DC x TCMS) Added Requirements: HR329, HR330, HR331, HR332 (overvoltage 1.5kV) , HR333, .HR334, (Master/slave Condition in AC) HR335, (panto up status in AC x TCMS) HR336,.HR337(undervoltage level 1 in 3kV) Renamed Req .543 as .HR340
4.4	R. Schiavo	16/10/2024	Mod req .HR175
4.5	R. Schiavo	15/11/2024	Mod req .HR276, .HR277
4.6	R. Schiavo	12/12/2024	Mod req .222,.HR281, .HR267

4.7	R. Schiavo	09/01/2025	Modified Req .1105, .731, .HR59, .HR68, .HR69, .1069 Renamed double Req .HR178 as .HR341 Deleted .HR248 Added req.HR342,.HR343,.HR344,.HR345
4.00	R. Schiavo	16/01/2025	Emission – BL3 baseline
5.1	P. Buda	30/05/2025	Added new requirements: .HR346,.HR347,.HR348,.HR349,.HR350 Modified the req. .1091
5.2	P. Buda	30/06/2025	Modified req: .1090, .1091. Deleted req: .1000
5.3	P. Buda	09/07/2025	Modified req: .HR71, .765, .497, .498
5.4	P. Buda	17/07/2025	Modified req: .HR03
5.5	P. Buda	21/07/2025	Added in par. 4.2, Propulsion Control -> CCU, “Status: AC Earthing switch is faulty”, req .HR351
5.6	P. Buda	25/07/2025	Modified req .HR343 Added req .HR352, .HR353
5.0	P. Buda	25/07/2025	Emission BL4 baseline

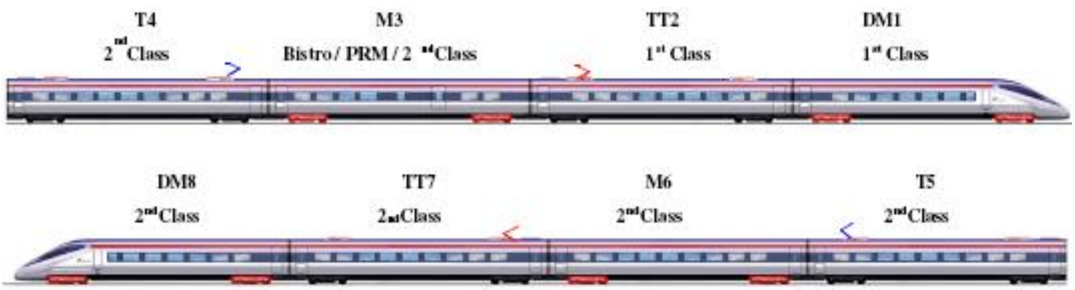
Content

1 Introduction

This document is a function specification for the control of the *Line Voltage system* for ETR1000 for customer Trenitalia. (VZI-6 Project).

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

AC	Alternative Current
ADD	Automatic Dropping Device
ATP	Automatic Train Protection
CD	Circuit Diagram
DC	Direct Current
DCU	Drive Control Unit
DCUL	Drive Control Unit / Line
DI-signal	Digital Input signal
DO-signal	Digital Output signal
ETCS System	European Train Control System
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
FMECA	Failure Mode Effect and Criticality Analysis
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
HV	High Voltage
HVSP	High Voltage System Protection
HW	Hardware
IDU	Interface Display Unit
KPS	Relay energized by pressure switch of the pantograph
LCB	Line Circuit Breaker
LCM	Line Converter Module
MCM	Motor Converter Module
MVB	Multifunction Vehicle Bus
N/A	Not Applicable
PCU	Propulsion Control Unit
PCU	Pantograph Control Unit
PPC	Propulsion & Controls
PPU	Pantograph Pneumatic Unit
SFRS	System Function Requirement Specification
SIL	Safety Integrity Level

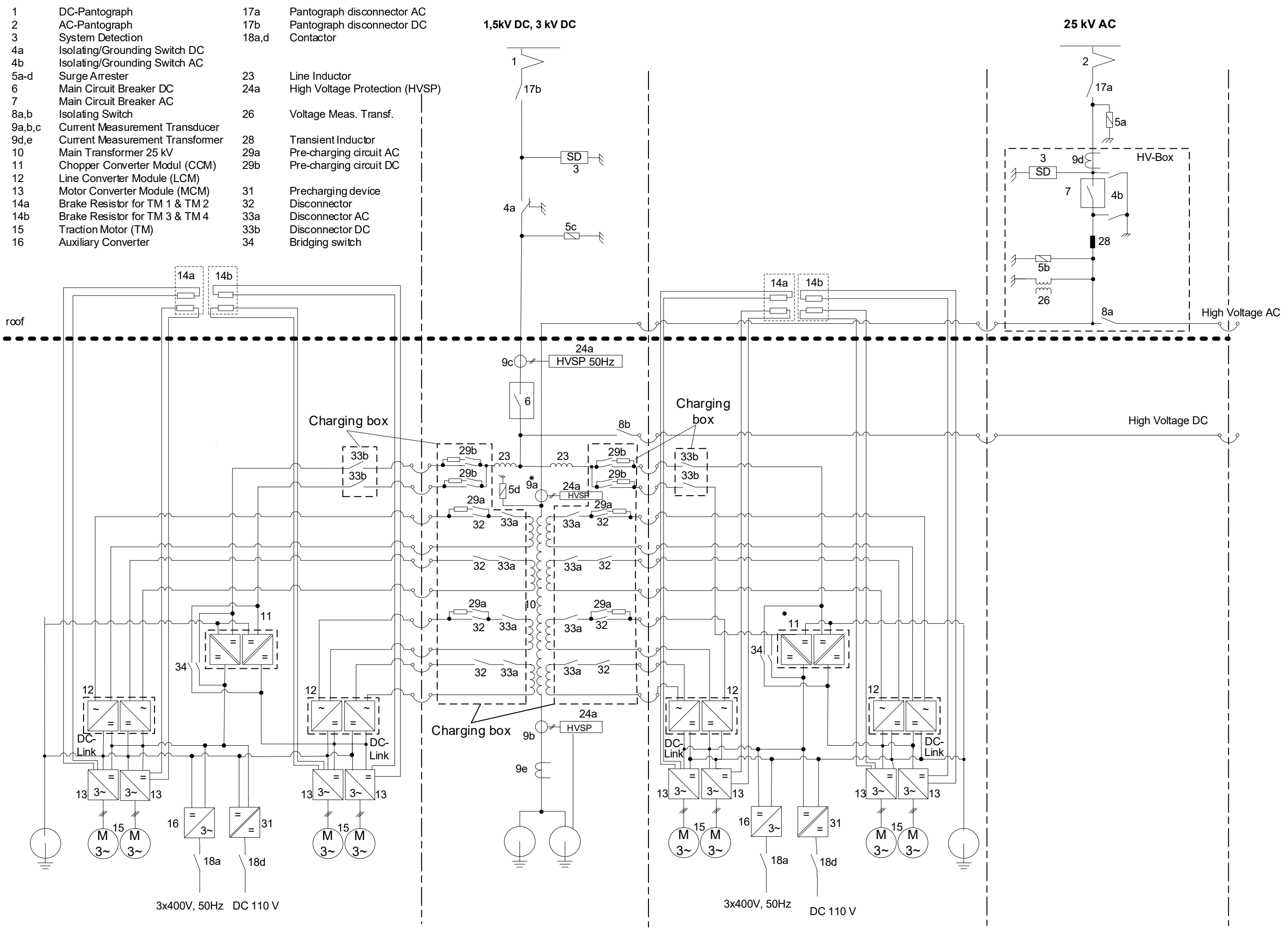
SIS	Safety instrumented system
SW	Software
TBD	To Be Defined
TCMS	Train Control and Management System
TRS	Technical Requirement Specification
VFRS	Vehicle Functional Requirement Specification

All numbers in brackets refer to related numbers of system overview (see section 2.1).

2 System Overview

2.1 System Configuration

The architecture of the V300ZEFIRO energy distribution and traction/el. braking systems are designed with two Traction Units. Each traction unit consists of 4 coaches (half train), mirroring each other.



ID	2F_04.01.Zefiro-Europe.TRS.670
Description	The architecture of the power supply system shall support the following three different line voltage systems: AC 25 kV 50 Hz, AC 15 kV 16.7 Hz, 3 kV DC to 1.5 kV DC.
Safety level	N/A

The architecture of the power supply system supports the following three different line voltage systems: AC 25 kV 50 Hz, 3 kV DC to 1.5 kV DC.

The line voltage system is selected by the train driver (see chapter 3.3.5 "system selection"). To identify the country of operation, the driver has to select the country manually.
The High voltage system is divided into two sections: AC high voltage section and DC high voltage section. Both parts are built-up redundantly, that means they consist of two pantographs for each catenary system and identical subsystems. Both line voltage

subsystems are connected via separate high voltage cables, thus each of the two pantographs can supply the whole trainset simultaneously. All pantographs and the system detection are able to withstand 25 kV AC.

3 Functional requirements

3.1 Country code selection

3.1.1 Function Design

3.1.1.1 *Normal condition*

One of the codes reported can be selected. Only Italy country code is operative on the VZI-6 trains

Code	Country of operation	Code Feedback
1	Italy	Ok
2	Austria	Not Available
3	Belgium	Not Available
4	France	Not Available
5	Germany	Not Available
6	Netherlands	Not Available
7	Spain	Not Available
8	Switzerland	Not Available
0		Not Defined

3.1.1.2 *Failure condition*

1) a Country Code not available is selected
2) A country Code not defined is selected

3.1.1.3 *Failure supervision*

1) If a country code not available is selected a message shall be raised to the driver
2) If a country code not defined is selected a message shall be raised to the driver

3.1.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.772
Description	It shall be possible to select the country which the train will be operated in from the TOD. Possible countries are: – Italy, Austria, Belgium, France, Germany, Netherlands, Spain and Switzerland. Only Italy code is operative The country selection shall be used for country specific configurations such as: – selection of wide or narrow pantograph – selection of Voltage system – levels of line voltage supervision – levels of line current harmonics.
Safety level	SIL 2

Nr	**Description**	**Applicability Details**	**Derived to**	**User Interface**	**SIL**
2F_04.01.Zefiro-Europe.CONCEPT.775	One of the following country-codes shall be used by TCMS depending the country in which the vehicle will operate: Code / Country of operation 1 / Italy 2 / Austria 3 / Belgium 4 / France 5 / Germany	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 2

	6 Netherlands 7 Spain 8 Switzerland 0 not defined				
2F_04.01.Zefiro-Europe.CONCEPT.965	DELETED.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1013	The selected country code shall be sent to all PCU's in the train.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.HR01	The selected country code shall be sent to HVSPs in the train.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR310	The country code shall be validated if equal to 1 (Italy code) and check variable = 1.	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.797	In emergency operation mode (emegency operation mode trainline is energised) the latest selected/known country code shall be used and communicated to the PCU .	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_04.01.Zefiro-Europe.CONCEPT.HR02	In emergency operation mode (emegency operation mode trainline is energised) the latest selected/known country code shall be used and communicated to the HVSP.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR03	It shall not be allowed to select a not available or not defined country code. Note: (Not applicable to VZI-6)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	

3.2 Pantograph Control

3.2.1 +Pantograph - Lifting

3.2.1.1 *Function Design*

3.2.1.1.1 Normal Condition

Lifting the pantograph is a precondition for collecting energy from catenary system.
Lifting can be performed automatically or manually. In any case, the driver selects a line voltage system (4-position selector switch) and switches "raise pantograph 1" or "raise pantograph 2".
Country Code is needed to identify pantograph which is needed depending on country of operation.
The identified pantograph is lifted up on request.
TCMS sends to PCU(Pantograph Control) by IP bus (TRDP protocol) which type of pantograph has to be lifted. TCMS also commands 2 different DOs in order to define which type of pantograph has to be lifted.

3.2.1.1.2 Failure Condition

1) Pantograph doesn't lift up although an order was given.
2) Two pantographs of the selected type are lifting up. (unless for ice scraping)
3) Lifting of pantograph without order.
4) Wrong type of pantograph is lifting up.
5) Pantograph switch is in "up" position although related pantograph is lowered.

3.2.1.1.3 Failure Supervision

1) If the pantograph is not raised after 35 seconds after the raise order and the main reservoir pressure is higher than 750 kPa, the pantograph is considered faulty.

If no air pressure is detected at the pantograph 5 minutes after the raise order and the main reservoir pressure is less than or equal to 750 kPa, the pantograph is considered faulty.

2) Both pantographs will be ordered down.
3) Pantograph will be lowered, considered faulty and blocked additionally an event will be set.
4) system detection detects invalid line voltage system or country code does not fit to raised pantograph. In both cases all pantographs will be lowered and a failure message will be displayed.
5) TCMS ignores pantograph-switch and pantograph remains in lowered position. IDU indicates that switch has to be turned to "down" position.

3.2.1.2 *Requirements*

ID	2F_04.01.Zefiro-Europe.TRS.619
Description	The following sequence shall be implemented: * Driver selects the Country where the train is operated from the IDU. * Driver selects the line voltage system * The (not blocked or cut out) pantograph (one per 8 car train unit) for the chosen line voltage system shall be lifted when requested by the driver * Line voltage shall be detected and the chosen line voltage system verified If the detection and verification are successful, the closing of line circuit breakers shall be enabled. ELSE a diagnostic event shall be generated.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.735	With input from train driver (line voltage selector, see chapter 3.3.5 "System Selection") and depending on country of operation (country-code), TCMS shall select the needed type of pantograph (see chapter 3.2.5 "Pantograph selection").	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1099	If a pantograph control unit (PCU) is faulty an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1096	In case all pantograph control units (PCU) are faulty, TCMS shall communicate directly to pneumatic control unit (PPU) which pantograph shall be ordered to raise. The train speed shall be limited to 160 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1113	The backup raise order to the pneumatic control unit (PPU) unit shall be connected to a TCMS DO. When the DO is energized the EV1 (EV2) order shall be notified to the PCU via IP.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1114	The major failure indication from PCU shall be connected to a TCMS DI.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1168	If Major Fault from PCU has been detected in TCMS while the corresponding pantograph is raised, then "LCB emergency off" train line shall be immediately energized and the raising command shall be removed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1174	Diagnosis of the contact: Major Fault (PCU) During the start up phase, after self test of PCU, when the life signal from PCU is being received, this test begins and TCMS shall manage it: – TCMS reads the contact(Digital input): Major Failure in closed position (i.e no failure inside PCU); – TCMS sends for 6 s to PCU the command to open (just for test) the contact: Major Failure (via bus).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

	This test is being carried out only if the pantograph is in lowered position. When the "Major Fault" contact test is on going, if the "Raise pantograph" switch is active, the raise order should be delayed to the end of the "Major Fault" contact test.				
2F_04.01.Zefiro-Europe.CONCEPT.1175	After self test, when PCU receives from TCMS the demand to carry out the test of the contact: "Major Fault", it has to follow exactly the request from TCMS in order to sincronize this test. PCU opens its own "Major Fault" contact for 2 s for test purpose.	Derived Requirement	4S_04.01.02.-.Pantographs		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1185	While TCMS is carrying out the test of Major Fault contact (i.e. TCMS is sending to PCU the request to open that contact): – TCMS reads the the contact: Major Failure in open position (i.e. it is not stuck); – if the contact doesn't open during the test, at the end of this test an event shall be set and the affected pantograph shall be considered as "Faulty lowered"	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1237	When major failure test is performed the TCMS shall rise an event for engineering of 'major fault test ongoing'	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1236	The major failure contact test has shall be done also when master reset is ordered	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1097	If the pantograph control unit rises the variable 'not rised on command' an event shall be raised to the driver	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1156	When external power supply is active the pantograph shall not be allowed to raise.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.767
Description	If the "Pantograph 1" switch is turned to "up" position the front pantograph in the train shall be raised.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.980	The "up" position of the Pantograph 1 switch shall be connected to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.544	When driver activates "Raise pantograph 1" switch, the front pantograph of the needed pantograph type shall be raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_04.01.Zefiro-Europe.TRS.768
Description	If the "Pantograph 2" switch is turned to "up" position the rear pantograph in the train shall be raised.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.982	The "up" position of the Pantograph 2 switch shall be connected to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.736	When driver activates "Raise pantograph 2" switch, the rear pantograph of the needed pantograph type shall be raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_04.01.Zefiro-Europe.TRS.685
Description	For speeds greater than or equal to 300 km / h, the maximum number of pantographs raised shall be limited to two per train.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1058	In case front pantograph is raised and selector for rear pantograph is activated, rear pantograph shall not be allowed to raise.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1043	An event shall be set if front pantograph is raised and selector for rear pantograph is activated stating that “Ice scraping is only allowed if rear pantograph is raised first”.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.793	In multiple operation, when driver activates "Raise pantograph 1" switch, in both coupled trains the front pantograph of the needed pantograph type shall be raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1002	In a multiple coupled train: If pantograph 1 is selected to raise then the following actions shall be performed in priority order: 1) If both front pantographs are not ok then no pantograph shall be raised and an event shall be set for driver to change pantograph. 2) If front pantograph of first train is ok then front pantograph of first train shall be raised. 3) If front pantograph of second train is ok then front pantograph of second train shall be raised. 4) If front pantograph of first train is not ok and rear pantograph of first train is ok then rear pantograph of first train shall be raised. 5) If front pantograph of second train is not ok and rear pantograph of second train is ok then rear pantograph of second train shall be raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.794	In multiple operation, when driver activates "Raise pantograph 2" switch, in both coupled trains the rear pantograph of the needed pantograph type shall be raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1003	In a multiple coupled train: If pantograph 2 is selected to raise then the following actions shall be performed in priority order: 1) If both rear pantographs are not ok then no pantograph shall be raised and an event shall be set for driver to change pantograph. 2) If rear pantograph of first train is ok then rear pantograph of first train shall be raised. 3) If rear pantograph of second train is ok then rear pantograph of second train shall be raised. 4) If rear pantograph of first train is not ok and front pantograph of first train is ok then front pantograph of first train shall be raised. 5) If rear pantograph of second train is not ok and front pantograph of second train is ok then front pantograph of second train shall be raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1004	If in DC operation rear pantograph of first train and front pantograph of second train are raised there shall be a speed restriction of 300 kph (fine tuning to be done later) and an event for driver to change	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

	pantograph “Speed restriction due to pantograph fault. Change pantograph if high speed on DC line is required”				
2F_04.01.Zefiro-Europe.CONCEPT.768	When an AC pantograph is not allowed to raise because there is already one up, this shall be indicated on IDU as an event.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.782	When any local pantograph-switch is in 'up' position at the cab activation (Note: not during parking exit procedure) an event to the driver shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.274	The related LCB (6,7) and system isolating switch (4a only in DC line voltage system) and AC earthing switch (4b only in AC line voltage system) must be in open position to lift the pantograph.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.563	When the pantograph is requested to raise, the pantograph disconnector (17) shall be closed first. When the feedback that the disconnector is closed is present, then the panto shall be ordered to raise.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1072	The pantograph disconnector shall close when ordered by TCMS DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1159	Pantograph control unit (PCU) commands to lift the selected pantograph, only if there is congruence between information coming via IP bus and signals from digital input.	Derived Requirement	4S_04.01.02.-.Pantographs		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.281	In order the keep the pantograph up the pantograph main control valve shall be kept open by energizing the solenoid.	Derived Requirement	4S_04.01.02.-.Pantographs		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1173	In case the IP communication between Pantograph Control Unit (PCU) and TCMS has been missed, PCU is not allowed to block and the pantograph mustn't be lowered.	Derived Requirement	4S_04.01.02.-.Pantographs		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1115	The pantograph control unit (PCU) receives 2 different raise orders depending from the different type of pantograph. Each one shall be connected to a dedicated TCMS DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.542	If the air pressure to the pantograph is low (pressure switch) the auxliliary compressor feeding the selected pantograph shall be ordered to start when the order to raise the pantograph is given.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.543	If the air pressure to the pantograph is low (pressure switch) AND the main recevoir pressure is <6 bar; the order to raise the pantograph shall be set when: – pressure switch toggles to 'pressure available' OR – indipendently from the pressure switch after 6 minutes that the auxiliary compressor has been running continuosly.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR340	In case the pantograph rise order (at the first rising) is delivered based on the condition: ' after 6 minutes that the auxiliary compressor has been running continuosly' an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.283	The raise order needs to be continuously high to raise and keep the pantograph in raised position. A raise order **must not be cancelled within the first 2 seconds** after	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

	activating it, else the pantograph structure might be harmed.				
2F_04.01.Zefiro-Europe.CONCEPT.1160	When DC line voltage system is selected and parking mode is activated, the second pantograph of same type has to be raised by TCMS without command from driver, but no change in LCB position. When AC line voltage system is selected and parking mode is activated, only the pantograph with the related LCB closed shall be kept raised	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.284	The pantograph status, with following priority order: – which pantos are "Cut out" – which pantos are "Faulty raised" – which pantos are "Faulty lowered" – which pantos are "Raised" – which pantos are "Lowered" shall be displayed on the IDU	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_04.01.Zefiro-Europe.CONCEPT.749	When a pantograph is raised(but not for ice scraping), the information which pantograph is raised shall be transmitted to propusion system to enable activation of related high voltage equipment.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR04	When a pantograph is raised (but not for ice scraping), the information which pantograph is raised shall be transmitted to HVSP	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_04.01.Zefiro-Europe.CONCEPT.HR226	If DC System Voltage is selected TCMS defines the pantograph status up if the PCU sends the pantograph raised feedback and the Voltage value from HVSP is > 800V	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_04.01.Zefiro-Europe.CONCEPT.HR335	If AC System Voltage is selected TCMS defines the pantograph status up if the PCU sends the pantograph raised feedback and the Voltage value from HVSP is > 3kV	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_04.01.Zefiro-Europe.CONCEPT.737	When pantograph is raised, present line voltage system shall be detected by system detection (see chapter 3.3.5.2 "System selection").	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1107	For any pantograph configuration in multiple operation (AC or DC) a speed limitation has to be implemented. For now the limit will be equal to the max. test speed (final tuning will be done after first tests): • 396 kph in AC 25kV • 330 kph in AC 15kV330 kph in DC 3kV • 242 kph in DC 1.5kV	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.304
Description	To free the catenary line from ice it shall be possible to lift two pantographs per 8car train, however only one line circuit breaker is closed to supply the train with energy. In this case it is necessary to foresee a speed limitation.
Safety level	SIL 0

To free the catenary line from ice it will be possible to lift two pantographs per 8-car train, however only one line circuit breaker is closed to supply the train with energy.
In this case it is necessary to foresee a speed limitation.

Nr	Description	Applicability Details	Derived to	User Interface	SIL

2F_04.01.Zefiro-Europe.CONCEPT.783	For ice scraping it shall be allowed to turn both pantograph switches to "up" position when the train is operating in Italy (country-code = 1) or France or Spain.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1238	When external temperature is <3°C **AND** train is running (V> 3km/h) **AND** the line selector is on 3kV **OR** 1,5kV a message 'risk of frozen catenary line' shall be raised to the driver after a time delay of 10 sec. the message shall be reset in the following cases: Ice scraping mode has been activated **OR** line selector set on AC mode **OR** external temperature is >=5°C for more than 10 sec	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.784	When both pantographs are raised for ice-scraping, the enable signal to close the LCB shall only be given to LCB which is related to the rear pantograph!	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.559	For ice scraping the pantograph closest to the active cab shall be used. NOTE! No ice scraping pantograph shall be used in a coupled train nor in backup mode.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.560	Ice scraping mode shall only be allowed in Italy, France and Spain.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

When a second pantograph is raised for ice scraping, it is in responsibility of the driver to limit the maximum speed at high speed lines to 200 km/h.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.548	If two DC pantographs is raised for ice scraping, a warning message shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.484	The status of the ice scraping pantograph shall be displayed on the IDU in the same way as for a normal power collecting pantograph	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.486	If a second pantograph is raised (for ice scraping) then the LCB and the System Isolation Switch (only in DC) connected to the second pantograph raised shall be interlocked.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.748	If a second pantograph is raised (for ice scraping) then the LCB and the System Isolation Switch (only in DC) connected to the second pantograph raised shall be interlocked by software.	Derived Requirement	4S_04.-.Power Supply		SIL 0

3.2.2 Pantograph - Lowering

3.2.2.1 *Function Design*

3.2.2.1.1 Normal Condition

Lowering of pantograph, the Pantograph will stop collecting line current.
Pantograph will be lowered when line voltage system is being de-activated:

- by switch “Pantograph down” OR
- by drivers urgency (“Emergency push button”) push button OR
- by automatic device due to system change OR
- in case of critical failures (Pantograph control unit output “pantograph internal major failure” in case of Pantograph control unit fault or difference in pressure control)

– at cab deactivation (no parking mode active)

All line circuit breakers are ordered to open before lowering the pantograph. except in emergency cases.

3.2.2.1.2 Failure Condition

1) Pantograph doesn't lower on order.
2) Pantograph is lowered unintentionally.

3.2.2.1.3 Failure Supervision

1) Manual lowering via panto-cock.
2) The faulty pantograph will be blocked.

3.2.2.2 *Requirements*

ID	2F_04.01.Zefiro-Europe.TRS.266
Description	Pantograph shall be lowered when line voltage system shall be de-activated – by switching the selector switch to the down position OR – by drivers emergency push button OR – in case of failures. All line circuit breakers shall be open before lowering the pantograph except in emergency cases.
Safety level	SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.769
Description	If the "Pantograph 1" switch is turned to "down" position the front pantograph in the train shall be lowered.
Safety level	N/A

ID	2F_04.01.Zefiro-Europe.TRS.770
Description	If the "Pantograph 2" switch is turned to "down" position the rear pantograph in the train shall be lowered.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.984	The "down" position of the Pantograph 1 switch shall be connected to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.983	The "down" position of the Pantograph 2 switch shall be connected to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.968	When driver activates "Lower pantograph 1" switch, the front pantograph of the needed pantograph type shall be lowered.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.969	When driver activates "Lower pantograph 2" switch, the rear pantograph of the needed pantograph type shall be lowered.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.561	When panto down order is given, the pantograph shall lower as soon as the LCB has opened or latest 3 seconds after the order was given.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1219	When a panto lowering order is given, a new rise command to the same pantograph shall be ignored untill the PCU delivers the information of panto lowered (Ipanpos = 0).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1220	In case a panto rise order is requested from the driver while the pantograph moving signal is set (during the lowering phase) an advisory should be displaied to driver asking to wait for the lowering phase to finish.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1225	When the pantograph is lowered (Ipanpos = 0), and the driver has tried to re-rise it during the lowering phase, an event shall be set to tell the driver that the panto is risable again.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

2F_04.01.Zefiro-Europe.CONCEPT.370	The pantograph shall not lower, except for specific emergency situations (stated in other requirements) before to reset the traction request and the LCB enable closing command .	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.565	The pantograph disconnector shall be ordered to open after the pantograph has been lowered or at latest 10 seconds after the pantograph down order was given.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1073	The pantograph disconnector shall open when ordered by TCMS DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.77	In order to lower the pantograph the pantograph main control valve shall be switched over to drain the air pipes by removing the power to the solenoid.	Derived Requirement	4S_04.01.02.-.Pantographs		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.289	When lowering a DC pantograph the system isolation switch shall open when the pantograph is in down position.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.562	The pantograph shall be order to lower 500ms after an emergency stop button is pressed, regardless of the position of the LCB.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1098	The pantograph shall lower, using the fast lowering valve (panto fast down train line), 500ms after an emergency stop button is pressed, regardless of the position of the LCB.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.994	Any raised pantograph shall be lowered when requested from the PCU(propulsion).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1008	If the cab is deactivated without entering the parking mode all pantographs shall be lowered.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1153	In case pantograph is lowered without order (signal from PCU) the LCB shall be opened by activation of "LCB emergency open" trainline, the pantograph order shall be removed and an alarm shall be set accordign to the diagnosys from the PCU(Pantograph Control Unit).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1165	In case Pantograph Control unit (PCU) detects incongruence between HW and SW 'raising signal' (= SW rise order is missing OR HW rise order is missing), the following actions shall be carried out. After 2 s from inchoerence detection: – **Predisposition**: LCB shall be ordered to open (TCMS via SW); – **Predisposition**: the pantograph shall be lowered and blocked – **Predisposition**: an event shall be set to the driver (Change pantograph required) – the related fault event shall be set to the maintainer (incoherence between HW and SW orders or vice-versa) After 16 s from inchoerence detection (if the pantograph has not been lowered): – **Predisposition**: TCMS shall energize "Panto Fast down " train line;	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1172	If the Pantograph Control Unit (of the panto in use) communication to the TCMS is NOT OK the following sequence shall be ordered: – LCB emergency open trainline shall be energized;	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

	– after 500 ms, Panto Fast down trainline shall be energized and and the raising command shall be removed;				

3.2.3 Automatic Dropping Device (ADD)

3.2.3.1 *Function Design*

3.2.3.1.1 Normal Condition

Lowering pantograph in case of collector head failure.

In order to have information about strips wearing there are two tubes inside each contact strip that are connected with pneumatic system, the tube at the highest level provides information about the wear of contact strips. The second tube at the lowest level activates the automatic lowering of pantograph (ADD).

When the ADD detects damages of contact strips it will drop the pantograph immediately.
In the event of collector head failure all pantographs of the train will be lowered immediately. After activation of ADD it must be possible to lift faultless pantographs.

3.2.3.1.2 Failure Condition

1) Strip is faulty but not detected by ADD.
2) Strip is not faulty but failure is detected by ADD.

3.2.3.1.3 Failure Supervision

1) No error-handling possible.
2) A faulty ADD-release will be handled as a normal ADD-release.

3.2.3.2 *Requirements*

ID	2F_04.01.Zefiro-Europe.TRS.445
Description	The ADD shall detect damages of contact strips (rupture, abnormal wear) and insulated end horns.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.332	The ADD shall drop the pantograph immediately if any damage to the contact strips are detected.	Derived Requirement	4S_04.01.02.-.Pantographs		SIL 2

ID	2F_04.01.Zefiro-Europe.TRS.270
Description	In the event of collector head failure all pantographs of the train shall be lowered immediately. After activation of ADD it must be possible to lift faultless pantographs.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1061	When wear detection is activated, the Pantograph control unit (PCU) shall communicate to TCMS that a contact strip is worn.	Derived Requirement	4S_04.01.02.-.Pantographs		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1062	If wear detection is tripped an EVENT shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1063	If wear detection is tripped a counter shall be activated to measure the running distance with the worn pantograph in use.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_04.01.Zefiro-Europe.CONCEPT.1119	If worn pantograph is in use, TCMS shall send to PCU (pantograph control unit) the process data “Wear detection blocked”.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1064	The counter shall be allowed to reset to 0 by maintenance people when worn contact strip has been changed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.82	If ADD is tripped the status for the pantograph shown on the IDU shall be "faulty".	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.294	If ADD is tripped an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.295	If ADD is tripped the LCBs shall open immediately.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.747	If ADD is tripped the LCBs shall open immediately.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.371	If ADD is tripped the LCBs shall be requested to open immediately.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1178	If ADD is tripped and the LCB Emergency Off activation is not read in every line trip chain of whole composition an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1169	In case ADD has been detected, via the KPD relay, the raising command from TCMS has to be removed	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1170	Pantograph Control unit (PCU) resets the signal: "ADD tripped" after 30 s that the pantograph has been folded.	Derived Requirement	4S_04.01.02.-.Pantographs		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1171	The condition: "ADD tripped" is being reset in TCMS as soon as the signal (relay KPD) has gone to 0	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1067	In case of ADD tripped the pantograph has to be declared as faulty and the panto has to be blocked. The panto will be operational again only after a master reset.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1110	A block of pantograph due to ADD-trip shall be allowed to reset only by maintenance people.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1068	If a faulty pantograph is blocked (due to ADD-trip), an EVENT shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.297	If ADD trips and train speed is > 0 km/h every pantograph in the whole train shall be lowered immediately via the "panto fast down" trainline.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1166	If "Panto Fast down" trainline is energized,TCMS sends this information to Pantograph Control Unit (PCU) via IP	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.296	If ADD trips and train speed is > 0 km/h every LCB shall requested to be opened and all pantographs in the whole train shall be lowered.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.372	The "fast panto down" trainline shall be set for 30 seconds when a ADD trip occurs.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1094	When pantograph raise order is given, activation of "fast panto down" trainline shall be delayed by n seconds to avoid that all pantographs will be lowered immediately.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.298	If the carbon strip is faulty the pneumatic control loop for the pantograph shall be drained.	Derived Requirement	4S_04.01.02.-.Pantographs		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.300	Insulating distance shall be reached immediately (worst case after 1s).	Derived Requirement	4S_04.01.02.-.Pantographs		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.301	After all pantographs have been dropped by ADD activating the "panto fast down" trainline, a non faulty pantograph shall be possible to select by the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_04.01.Zefiro-Europe.CONCEPT.1176	To avoid false diagnosis, PCU has to acquire the status of EV3 (inside PPU) and consequently to mask the diagnostic signal "ADD tripped" if EV3 has been energized	Derived Requirement	4S_04.01.02.-.Pantographs		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1157	In case of failure of relays in ADD detection circuitry (unavailability for LCB opening and panto fast down train lines activation), the raising command for the pantograph shall be blocked and an EVENT shall be set. Note: this diagnosis has to be masked in case of "Panto Fast down" train line has been energized or any A-Key has been activated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1158	Even though the pantograph was blocked due to failure recognised in ADD detection circuitry, in emergency operation mode the pantograph raising command shall be allowed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.794
Description	In case of unavailability of both pantographs of the available system due to ADD intervention (or failure), through a sealed switch in driver's cab the driver, under his responsibility, can allow to raise the pantograph and move the train at reduced speed in order to reach a siding track to clear the main line.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1291	GIVEN driver has switched the ADD bypass switch to AC or DC position AND the TCMS has accordingly activated the related DO of the ADD bypass solenoid valve THEN this solenoid valve prevent the leakage of air pressure due to any damage of contact strip on those pantograph types (AC or DC)	Derived Requirement	4S_04.01.02.-.Pantographs		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1292	GIVEN the driver has switched the ADD bypass switch to DC position THEN CTC shall forward the status DC ADD bypass active to TCMS via safe RIO (DI) Note:- DC ADD bypass means the ADD function of the DC pantograph is bypassed	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1293	GIVEN the driver has switched the ADD bypass switch to AC position THEN CTC shall forward the status AC ADD bypass active to TCMS via safe RIO (DI) Note:- AC ADD bypass means the ADD function of the AC pantograph is bypassed	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1339	ADD bypass switch is in "0", "DC" and "AC" according concept 1340	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1341	Event related to ADD bypass switch according concept 1340	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1340		Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

INPUT					OUTPUT		
MIO safe DC1	MIO safe DC2	MIO safe AC1	MIO safe AC2	Validity	Status ADD Bypass Switch	SET Event ADD Bypass Switch	RESET Event ADD Bypass Switch
0	0	0	0	1	0	0	1
1	1	0	0	1	DC	0	1
0	0	1	1	1	AC	0	1
1	0	X	X	X	0	1	0
0	1	X	X	X	0	1	0
X	X	1	0	X	0	1	0
X	X	0	1	X	0	1	0
1	1	1	1	X	0	1	0
0	0	0	0	0	0	1	0

Validity = 1 if all inputs (DC1, DC2, AC1, AC2) are valid
X = 1 OR 0

2F_04.01.Zefiro-Europe.CONCEPT.1295	GIVEN the ADD bypass switch in active driver's cab is in position "DC" OR the ADD bypass switch in active driver's cab is in position "AC" WHEN vehicle is at standstill THEN an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

2F_04.01.Zefiro-Europe.CONCEPT.1296	GIVEN the ADD bypass switch in active driver's cab is in position "DC" WHEN vehicle is at standstill THEN the solenoid valve for ADD bypass on both T2/T7 cars shall be energized by safe DOs	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1297	GIVEN the ADD bypass switch in active driver's cab is in position "AC" WHEN vehicle is at standstill THEN the solenoid valve for ADD bypass on both T4/T5 cars shall be energized by safe DOs	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1298	GIVEN local cab is not active WHEN the ADD bypass switch in local cab is in position "DC" OR the ADD bypass switch in local cab is in position "AC" THEN an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1307	GIVEN the event as for req#1298 has been set in this local cab WHEN the ADD bypass switch in local cab is not in position "DC" AND the ADD bypass switch in local cab is not in position "AC" THEN the event as for req#1298 for this local cab shall be reset	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1300	GIVEN driver has switched the ADD bypass switch to AC or DC position AND the TCMS has accordingly activated the safe DOs to energize in parallel, via diode, the KPS relay THEN the KPS relay usually fed by pressure switch (PS3 / PS4) is in position of ADD intervention not activated	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1301	*The feedback from pressure switch to indicate the ADD intervention (PS3 / PS4) shall be suppressed by TCMS in the following way*: GIVEN the ADD bypass switch in active driver's cab is in position "DC" THEN TCMS shall activate the safe DOs in order to energize the KPS relay related to ADD pressure switch (PS3 / PS4) , on both T2/T7 cars.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1302	*The feedback from pressure switch to indicate the ADD intervention (PS3 / PS4) shall be suppressed by TCMS in the following way*: GIVEN the ADD bypass switch in active driver's cab is in position "AC" THEN TCMS shall activate the safe DOs in order to energize the KPS relay related to ADD pressure switch (PS3 / PS4) , on both T4/T5 cars.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1303	GIVEN the ADD bypass switch in active driver's cab is in position "DC"	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

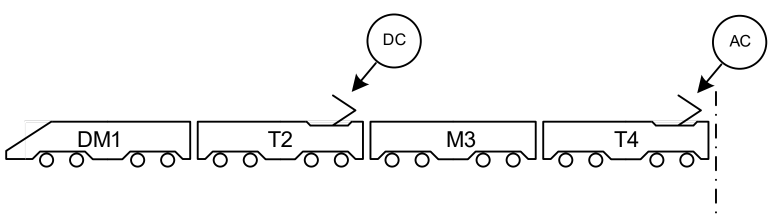
	THEN TCMS shall ignore any ADD intervention received from DC pantographs, in order to prevent any action which would bring to isolate the corresponding pantograph.				
2F_04.01.Zefiro-Europe.CONCEPT.1304	GIVEN the ADD bypass switch in active driver's cab is in position "AC" THEN TCMS shall ignore any ADD intervention received from AC pantographs, in order to prevent any action which would bring to isolate the corresponding pantograph.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1305	GIVEN pantograph has been declared as faulty and is blocked due to ADD intervention WHEN ADD bypass for that type of pantograph is active THEN panto blocking shall be temporarily suppressed and can be operational again	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1306	GIVEN panto blocking has been temporarily suppressed WHEN ADD bypass for that type of pantograph is no longer active THEN the pantograph remains blocked	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1333	GIVEN ADD bypass switch in active driver's cab is in position "DC" OR ADD bypass switch in active driver's cab is in position "AC" THEN Panto Control Unit shall receive information of "ADD Bypass Activated"	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.2.4 Pantograph - Interlock

3.2.4.1 *Function Design*

3.2.4.1.1 Normal Condition

Faulty pantographs are blocked automatically. The driver can block a pantograph manually to prevent switching or if it is defect without automatic detection.



3.2.4.1.2 Failure Condition

1) Unidentified lifting of pantograph.

3.2.4.1.3 Failure Supervision

1) Pantograph will be lowered and blocked.

3.2.4.2 *Requirements*

ID	2F_04.01.Zefiro-Europe.TRS.275
Description	The driver shall be able to manually block a pantograph.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.87	The driver shall be able to manually cut out a pantograph from selection	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_04.01.Zefiro-Europe.CONCEPT.303	If the pantograph is manually cut out an event shall be set as long as the pantograph is blocked.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.571	If a raised pantograph is manually cut out the LCB shall be requested to open and the pantograph shall lower.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.304	It shall not be possible to remove a manual cut out of the pantograph as long as any pantograph is raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.305	If a pantograph is blocked it shall always be considered down regardless of the pressure sensor indicates low or high pressure.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.306	If a pantograph is blocked the raise order shall be constant low.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_04.01.Zefiro-Europe.TRS.621
Description	A faulty pantograph shall automatically be blocked.
Safety level	SIL 0

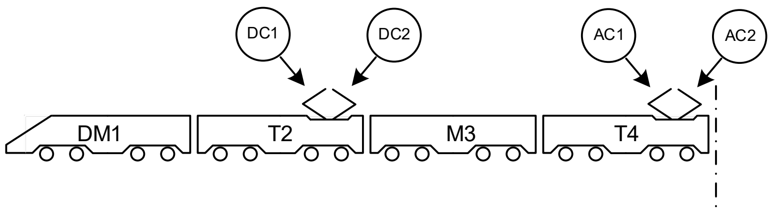
Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.309	In case the pantograph does not lower on command the diagnosys of the PCU shall rise an alarm to the driver and the pantograph shall be blocked.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.310	If the pantograph is in moving status (checking the pantograph position variable on IP) for t>20s an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.566	If the pantograph disconnector does not open when ordered the pantographs connected to that LCB (DC or AC) shall be permanently blocked and an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.567	If the pantograph disconnector does not close when ordered the pantograph shall be lowered, the pantographs connected to that LCB (DC or AC) shall be permanently blocked and an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1103	If the pantograph disconnector opens without order ("closed" feedback disappears) while the pantograph is raised, the LCB shall be opened, the pantograph shall be lowered, the pantographs connected to that LCB (DC or AC) shall be permanently blocked and an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1074	Both the positions open and closed of the pantograph disconnector shall be connected to TCMS DI.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.2.5 Pantograph Selection

3.2.5.1 *Function Design*

3.2.5.1.1 Normal Condition

Pantographs of different type (AC and DC) and different width (1450mm, 1600mm and 1950mm) can be used. Depending of the country the train operates in different width is required. Therefore, a country code to identify the current country of operations selected by the driver.



DC1 - 1450mm for DC-operation in Italy
DC2 - 1950mm for DC-operation in all countries but Italy
AC1- 1450mm or 1600mm for AC-operation in Italy, France, Belgium, Switzerland and Spain.
* 1450mm for operation in a corridor including Italy and Switzerland.
* 1600mm for operation in a corridor including Italy and Spain.
* For Belgium and France both 1450mm and 1600mm can be used.

AC2 - 1950mm for AC-operation in Germany, Austria, Netherlands and Spain.

On the VZI-6 trains only panto DC1 and AC1 are present.

3.2.5.1.2 Failure Condition

1) Wrong pantograph type is ordered to lift up.
2) More than one pantograph gets raise order.

3.2.5.1.3 Failure Supervision

1) Driver supervises pantograph selection.
2) With exception of raising a second pantograph (ice scraping) all LCBs will open immediately and all pantographs will be lowered, if - in one 8 car train - more than one pantograph is raised.

3.2.5.2 *Requirements*

ID	2F_04.01.Zefiro-Europe.TRS.286
Description	The 4 pantographs of one half-train (4 cars), which can be used for AC voltage of 25 kV 50 Hz or 15 kV 16.7 Hz and DC voltage of 3 kV or 1,5 kV, shall be interlocked against each other. Switch over between the systems shall be performed automatically onboard the train set while the train is in motion.
Safety level	SIL 0

Only one pantograph on a full train set (8-cars) is selected for raising. The only situation when two pantograph selected is when one is used for ice scraping, see section 3.2.1

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.786	When voltage system AC 25kV or AC 15kV is selected and the train operates in Italy, the AC-pantograph (AC1, 1600mm) shall be selected.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1329	When voltage system DC 3kV is selected and the train operates in Italy the DC1-pantograph shall be selected.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_04.01.Zefiro-Europe.CONCEPT.496	If the LCB connected to the selected pantograph is blocked, it shall be indicated on IDU and the driver will have to raise the other pantograph.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1095	When one of the following countries is selected all pantographs shall be blocked: Germany, Austria, Switzerland, Belgium, Spain.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.325
Description	It shall be possible, through appropriate forcing procedures, to collect energy from any type of catenary by using the other AC pantograph of the trainset for AC lines or the other DC pantograph of the trainset for DC lines. This is limited for special purpose (like testing) under specific and supervised conditions.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.216	It shall be possible to control each pantograph individually by test parameters in the software. These parameters shall only be possible to be manipulated by connecting directly to the PCU (Pantograph Control Unit) with appropriate commissioning tools such as PTU.	Derived Requirement	4S_04.01.02.-.Pantographs (PCU)		N/A

3.2.6 Pantograph - Control of Static Contact Force

3.2.6.1 *Function Design*

3.2.6.1.1 Normal Condition

The static force of the raised pantograph will be controlled depending on speed, travel direction, position of pantograph and voltage system.

3.2.6.1.2 Failure Condition

1) Control system is not working correctly.
2) Ethernet link is missing.

3.2.6.1.3 Failure Supervision

1) Switching to other pantographs.
2) Switching to other pantographs.

3.2.6.2 *Requirements*

ID	2F_04.01.Zefiro-Europe.TRS.454
Description	The pressure shall be a function of voltage system, speed, travel direction and position of pantograph
Safety level	N/A

The pressure is a function of voltage system, speed, travel direction and position of pantograph

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.97	The static contact force of the pantograph shall be regulated using the following parameters – Voltage system (AC or DC) – Train speed – Travel direction (Backwards and Forward) – Position of all raised pantographs in a train	Derived Requirement	4S_04.01.02.-.Pantographs		N/A

	– (counting from 1, closest to active cab Note: for a coupled consist the numbering continues) – applicability of ice-scraping mode				
2F_04.01.Zefiro-Europe.CONCEPT.1111	TCMS shall provide following parameters to pantograph control unit (PCU): – Voltage system (AC or DC) – Train speed – Travel direction (Backwards and Forward) – Position of all raised pantographs in a train – (counting from 1, closest to active cab Note: for a coupled consist the numbering continues) – applicability of ice-scraping mode	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.3 Supervision of Line Voltage and Line Current

3.3.1 Line Voltage Supervision

3.3.1.1 *Function Design*

3.3.1.1.1 Normal Condition

The supervision of the Line Voltage is done by the voltage sensors connected to the propulsion equipment. If the line voltage detection indicates a wrong line voltage system or a too high or too low Line Voltage, the active LCB will be opened respectively not closed.
Precondition:

- A country code is selected to identify the country of operation
- Voltage System is selected

3.3.1.1.2 Failure Condition

1) No signal from system detection AC or DC (3).
2) Incorrect measurement (implausible voltage value) in system detection AC or DC (3).
3) No signal from AC voltage transformer (26)
4) Incorrect measurement (implausible voltage value) in AC voltage detection (26).

3.3.1.1.3 Failure Supervision

1) Driver has to change pantograph.
2) Driver has to change pantograph.
3) Select Remote AC Voltage Measurement Transformer
4) Select Remote AC Voltage Measurement Transformer

3.3.1.2 *Requirements*

ID	2F_04.01.Zefiro-Europe.TRS.458
Description	Line voltage shall be supervised if it is within acceptable range to allow the line circuit breakers to close and to release full or reduced traction
Safety level	SIL 1

If "AC system detection" (inside HV-Box, T4 & T5 car) detects AC (50Hz) contact wire voltage, the voltage of AC-system will be measured in addition (25 kV).
If DC system detection (TT2 & TT7 car) detects DC contact wire voltage, the voltage of DC-system will be measured in addition (1.5 kV or 3 kV).

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.222	If 25kV line voltage is selected, the AC-LCB shall be allowed to close when the line voltage detection AC 50 Hz is in the right range: ≥17.5 kV and ≤29.0 kV, THEN HVSP Master closes the line trip and sets high the ACLineVOk signal. If 25kV voltage is selected AND the AC LCB is open AND the line trip is closed AND the AC line voltage detected is out of the right range for 1 sec THEN HVSP Master resets the ACLineVOk signal and opens the line trip If 25kV voltage is selected AND the line trip is open AND the AC line voltage detected is out of the right range THEN HVSP Master resets the ACLineVOk signal and keeps open the line trip	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR281	If 3kV voltage is selected, the DC-LCB shall be allowed to close when the line voltage detection is in the right range: ≥1.975kV and ≤4.2 kV, THEN HVSP Master closes the line trip and sets high the DCLineVOk signal. If 3kV voltage is selected AND the DC LCB is open AND the line trip is closed AND the DC line voltage detected is out of the right range for 1 sec THEN HVSP Master resets the DCLineVOk signal and opens the line trip If 3kV voltage is selected AND and the line trip is open and the DC line voltage detected is out of the right range THEN HVSP Master resets the DCLineVOk signal and keeps open the line trip	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR267	If 1,.5kV voltage is selected, the DC-LCB shall be allowed to close when the line voltage detection is in the right range: ≥1.0kV and <1.975 kV.HVSP Master closes the line trip and sets high the DCLineVOk signal. If 1.5kV voltage is selected AND the DC LCB is open and the line trip is closed and the DC line voltage detected is out of the right range for 1 sec THEN HVSP Master resets the DCLineVOk signal and opens the line trip If 1.5kV voltage is selected AND the line trip is open and the DC line voltage detected is out of the right range THEN HVSP Master resets the DCLineVOk signal and keeps open the line trip	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1216	The LCB closing request has to be delivered to Propulsion only if one of the following conditions has been satisfied for 3 seconds: – the line voltage detection AC 50 Hz is ≥17.5 kV and ≤29.0 kV – the line voltage detection DC 3kV is ≥1.975 kV and ≤4.2 kV, – the line voltage detection DC 1.5kV ≥1.0 kV and <1.975 kV	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR352	If the LCB is closed and the following conditions has been satisfied for 1 seconds: – the line voltage detection AC 50 Hz is <17 kV OR >31.1 kV		4S_09.03.05.-.TCMS Software		SIL 0

	– the line voltage detection DC 3kV is <1.8 kV OR >4.2 kV, - the line voltage detection DC 1.5kV <0.85 kV OR >2.0 kV The LCB closing request has to be reset and an Event has to be generated				
2F_04.01.Zefiro-Europe.CONCEPT.HR353	If the conditions reported in the req2F_04.01.Zefiro-Europe.CONCEPT.HR352 are satisfied then the event “line Voltage out of range” shall be generated		4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR333	In AC configuration if HVSP slave receives the pantograph up signal High AND detects a line voltage >3kV THEN becomes HVSP Master	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR280	In DC configuration if HVSP slave receives the pantograph up signal High AND detects a line voltage >700V THEN becomes HVSP Master	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR282	In DC configuration If HVSP Master receives the DC pantograph up signal low AND the line voltage is not present (<450V) THEN becomes HVSP slave.	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR334	In AC configuration HVSP reset the Mastership status when receives the pantograph up signal low and the line voltage is not present (<2kV)	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR14	In each half Train a PCU is Half Train Master configured	Derived Requirement	4S_04.-.Power Supply (PCU)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR15	IF in the local half train the panto is up to connect the high voltage to the train THEN the PCU Half Train Master configured is also HV Master configured	Derived Requirement	4S_04.-.Power Supply (PCU)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR18	IF country code and voltage system are selected and panto is up THEN HV Enable is sent to HVSP	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR07	IF country code and voltage system are selected ant panto is up THEN Hv Enable is sent to PCUs	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR06	IF the LCB closing is requested by driver desk lever AND range VoltageOk from HVSP HV Master AND VoltageTypeOk from PCU HV Master THEN EnableLCBclosing is sent to PCUs	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR314	The EnableLCBclosing signal is reset if a line trip open signal is received from HVSP or from a PCU not cutout	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR343	The ProtectiveShutDown signal is set if (a line trip open signal is received from HVSP or from a PCU not cutout) and (the parking mode is not activated and the neutral/separation section is not activated) or (for any LCB opening due to protection)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR08	IF LCB Enalble is received from TCMS THEN the PCU HV Master energizes the line trip	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR09	IF a valid country code is received AND the Voltage range is coherent, with the system voltage selected (ref. .222, .HR281, .HR267) AND HVEnable is received from TCMS AND trip conditions are not present THEN HVSP HV MASTER energizes its line trip relay	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR269	If HVSP detects the line trip energized AND the HVEnable is not received in 10 seconds set a diagnostic signal	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 0

2F_04.01.Zefiro-Europe.CONCEPT.HR212	IF HVSP detects the line trip energized AND the HVEnable is not received in 10 seconds an event is created	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR271	IF HVSP receives the HVEnable AND country code is not valid in 2 seconds set a diagnostic signal	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR268	IF HVSP receives the HVEnable AND country code is not valid in 2 seconds an event is created	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR272	IF HVSP detects the line voltage NOT in the right range AND the LCB Enable Closing is received set a diagnostic signal	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR214	IF HVSP detects the line voltage NOT in the right range AND the LCB Enable Closing is received an event is created	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR274	If HVSP detects the line trip energised AND commands the lineTrip to Close AND detects the LineTrip Feedback Low after 1 second set a diagnostic signal	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR218	If HVSP detects the line trip energised AND commands the lineTrip to Close AND detects the LineTrip Feedback Low after 1 second an event is created	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR275	If HVSP commands the LineTrip to Open AND detects LineTrip Feedback High set a diagnostic signal	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR219	If HVSP commands the LineTrip to Open AND detects LineTrip Feedback High an event is created	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR10	IF the line trip is energized AND trip conditions are not present THEN HVSP slave energizes its line trip relay	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR11	IF trip conditions are not present THEN the converter controls energize the line trip relays	Derived Requirement	4S_04.-.Power Supply (MOCUs)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1217	In case of requests to close the LCB with a measured line voltage not within the limits above an advisory shall be rised to the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.313	The DC-LCB shall be allowed to close when a DC the line voltage is detected AND when system isolating switch (4a) is closed!	Derived Requirement	4S_04.-.Power Supply (circuit)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.501	When DC line voltage system 3.0 kV OR 1.5kV is selected and the line voltage DC is detected the system isolating switch (4a) is allowed to close	Derived Requirement	4S_04.-.Power Supply (circuit)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR12	When DC line voltage system 3.0 kV OR 1.5kV is selected and the line voltage DC is detected AND HVEnable is received from TCMS THEN the PCU HVMASTER close the System isolating switch)	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR222	The SIS close order signal shall be set low as soon as the feedback SIS CLOSED goes high.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR223	The SIS open order signal shall be set low as soon as the feedback SIS OPEN goes high.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR224	The maximum time for close or open SIS order shall be 5 seconds.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.328
Description	When the maximum line voltage for the nominal 25kV system is exceeded; after 10 minutes operation at line voltage above 29kV or after 1 second at line voltage above 31 kV; the line circuit breaker shall be opened.

Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1082	If Line voltage AC 50 Hz is >32.0kV for ≥1 sec the line trip shall be open or kept open	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR170	If Line voltage AC 50 Hz is >32.0kV for ≥1 sec the Excessive line voltage event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR171	If the Excessive line voltage event occurs the LCB enable shall be reset and the pantograph lowered	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.315	If the AC LCB is closed and the Line voltage AC 50 Hz is >=31.1kV for ≥1 sec (AC overVoltage level 2) or >=29.1kV for ≥10 minutes sec (AC overVoltage level 1) HVSP Master shall open the line trip	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.578	If the AC LCB is closed and the Line voltage AC 50 Hz is >=31.1kV for ≥1 sec an event for AC overVoltage level 2 shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.760	If the AC LCB is closed and the Line voltage AC >=29.1kV for ≥10 minutes an event for AC overVoltage level 1 shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.319	The 10 minute clock for line voltage AC 50 Hz supervision shall have following behaviour: – line voltage >=29.1kV, clock will be counted-up – line voltage ≤29.0kV for at least 200ms or line trip or AC LCB is open, clock will be resetted – max. time is 10 minutes, min. value for clock is 0	Derived Requirement	4S_04.-.Power Supply (HVSP)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.316	If the DC LCB is closed and the Line voltage DC 3kV >4.1kV (operation in Italy: >4.2kV, operation in Belgium: >4.0kV) (DC overVoltage level 2) for ≥1 s or > 4.0kV for ≥5 minutes (DC overVoltage level 1) HVSP Master shall open the line trip.	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.580	If the DC LCB is closed and the Line voltage DC 3kV >4.2kV for ≥1 s an event for DC overVoltage level 2 shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR209	If the DC LCB is closed and the Line voltage DC > 4.0kV for ≥5 minutes an event for DC overVoltage level 1 shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.503	The 5 minute clock for line voltage DC 3 kV supervision shall have following behaviour: – line voltage >4.0kV, clock will be counted-up – line voltage ≤4.0kV for at least 300ms, clock will be counted-down – max. time is 5 minutes, min. value for clock is 0	Derived Requirement	4S_04.-.Power Supply (HVSP)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR329	If the DC LCB is closed and the Line voltage DC 1.5kV >2kV for ≥1 s (DC overVoltage level 2) or > 1.9kV for ≥5 minutes (DC overVoltage level 1) HVSDP Master shall open the line trip.	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2

2F_04.01.Zefiro-Europe.CONCEPT.HR330	If the DC LCB is closed and the Line voltage DC 1.5kV >2kV for ≥1 s an event for DC overVoltage level 2 shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR331	If the DC LCB is closed with 1.5 kV system selected and the Line voltage DC > 1.9kV for ≥5 minutes an event for DC overVoltage level 1 shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR332	The 5 minute clock for line voltage DC 1.5 kV supervision shall have following behaviour: – line voltage >1.9kV, clock will be counted-up – line voltage ≤1.9kV for at least 300ms, clock will be counted-down max. time is 5 minutes, min. value for clock is 0	Derived Requirement	4S_04.-.Power Supply (HVSP)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.317	If the AC LCB is closed and the Line voltage AC 50Hz is <17,0 kV for ≥1 sec HVSP Master shall open the line trip shall	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.582	If the AC LCB is closed and the Line voltage AC 50Hz is <17,0 kV for ≥1 sec an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR210	If the AC LCB is closed and the Line voltage AC 50Hz is>= 17kV and <19 kV for 120 sec HVSP Master shall open the line trip	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR211	If the AC LCB is closed and the Line voltage AC 50Hz is>= 17kV and <19 kV for 120 sec the LCB an event for AC undervoltage level 1 shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.318	If the DC LCB is closed and the Line voltage 3kV is <1.8 kV for ≥1 sec HVSP Master shall open the line trip	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.583	If the DC LCB is closed and the Line voltage 3kV is <1.8 kV for ≥1 sec the LCB shall be opened and an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR336	If the DC LCB is closed and the Line voltage 3kV is>1.8kV and <=2 kV for 120 sec HVSP Master shall open the line trip	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR337	If the DC LCB is closed and the Line voltage 3kV is is >1.8kV and <=2 kV for 120 sec an event for DC undervoltage level 1 shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.541	If the DC LCB is closed and the Line voltage 1.5kV is <0.85 kV for ≥1 sec HVSP Master shall open the line trip.	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.584	If the DC LCB is closed and the Line voltage 1.5kV is <0.85 kV for ≥1 sec an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR260	When the AC LCB is closed IF no signal OR implausible voltage value from Local AC voltage measurement transformer (26) to all PCUs in the half Train THEN the local AC voltage measurement is faulty	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR261	If local AC voltage measurement transformer is faulty THEN the remote voltage measurement transformer has to be set	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR262	If the AC voltage measurement transformer is faulty an Event is created	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)	EVENT	N/A

ID	2F_04.01.Zefiro-Europe.TRS.793
Description	The vehicle shall display the actual line voltage value to the driver.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.321	The line voltage shall be displayed on IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.987	If line voltage is not detected within 5 seconds from when pantograph was indicated as raised an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.3.2 Line Current Supervision

3.3.2.1 *Function Design*

3.3.2.1.1 Normal Condition

AC line current measurement is done by line current transformer (9d) placed in AC HV-box on T4 and T5 car. It is connected to the HVSP which is responsible for overcurrent detection.

Harmonic DC line current measurement is done by DC line current sensor (9c).

At each main transformer a current transformer for line converter control is installed.

3.3.2.1.2 Failure Condition

1) No detection of line current when line current is expected.
2) No detection of transformer current when transformer current is expected.
3) Detection of line current when line current is not expected.

3.3.2.1.3 Failure Supervision

1) Event will be set and driver has to change pantograph.
2) Depending on the situation the transformer will be isolated, bypassed or event will be set. Handled by internal propulsion supervision.
3) Same action as LCB does not open on order

3.3.2.2 *Requirements*

ID	2F_04.01.Zefiro-Europe.TRS.460
Description	At each main transformer there shall be a current transformer to measure the line current.
Safety level	SIL 1

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.106	The current shall be measured by a return current transformer on the primary winding at each main transformer for the differential protection Note: Only applicable for AC operation, , the measure and the protection shall be realized from HVSP Master and Slave	Derived Requirement	4S_04.-.Power Supply (HVSP)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.988	The transformer currents (input and return current) shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR276	If a differential current >50 Arms for 100ms within 1 sec from AC LCB closing a differential overcurrent is detected detected on HVSP MASTER If a differential current >50 Arms for 100ms within 1 sec from line trip closing a differential overcurrent is detected on HVSP Slave	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2

2F_04.01.Zefiro-Europe.CONCEPT.HR277	If a differential current >30 Arms for 100ms after 1 sec from AC LCB closing a differential overcurrent is detectedon HVSP MASTER If a differential current >30 Arms for 100ms after 1 sec from line trip closing a differential overcurrent is detected on HVSP Slave	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1129	If a **differential overcurrent** is detected, the line trip shall be opened and reclosing shall be blocked for 5 seconds.	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR181	If 2 events of **differential overcurrent** occur in 3 minutes, HVSP keeps the line trip open	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR182	If 2 events of **differential overcurrent** occur in 3 minutes, TCMS cut out (only in AC configuration) the half train where the protections are occurred	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR183	If 2 events of **differential overcurrent** occur, TCMS sends the AC roof disconnector opening request to PCUs	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR184	If 2 **differential overcurrent** events in 3 minutes occur, TCMS requires the pantograph change to the driver if the events regard the transformer on the half train where the pantograph is raised	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.758	If a **differential overcurrent** is detected an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR346	An event shal be set by TCMS if the sum of AC second higher current detected by the PCU Master of first half train and AC second higher current detected by the PCU Master of second half train differs from the AC current detected by HVSP of 50 A for ≥ 1 s.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.251
Description	Current measuring unit is used for line current measurement on high voltage side. The output signal shall be used for train protection (over currents, short circuits, differential currents)
Safety level	SIL 1

In AC line voltage system as well as in DC line voltage system there is a current measurement device installed in front of the LCB. The over current (OVC) detection for AC is handled by the HVSP and for DC by the LCB itself.

1°LCB opening (OVC) -> wait 5sec -> 1° LCB re-closing
2°LCB opening (OVC)-> wait 5 sec and open roof Disconnector-> 2°LCB re-closing
3°LCB opening(OVC)-> wait 5 sec and change pantograph->3°LCB re-closing (Last attempt)
4°LCB opening (OVC) -> high Voltage blocked

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1130	The line current shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR205	HVSP sends AC line current, Primary trafo current, DC line current 50Hz Harmonic, DC and AC line Voltage measured RMS values to TCMS	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR206	PCU sends DC line current for an half train as sum of 4 CCMs input current	Derived Requirement	4S_04.-.Power Supply (PCU)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.225	If a AC line current is > 1800A an AC line overcurrent is detected, HVSP Master shall open the line trip and reclosing shall be blocked for 5 seconds.	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 1
2F_04.01.Zefiro-Europe.CONCEPT.HR322	When a LCB opening without command occurs the enable to reclose shall be avoided for 5 seconds.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_04.01.Zefiro-Europe.CONCEPT.361	First attempt of reclosing LCB: If AC line overcurrent is detected occurs within 5 seconds after first re-closure of the LCB, the LCB shall be opened again.	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 1
2F_04.01.Zefiro-Europe.CONCEPT.HR16	IF two LCB opening due to AC line overcurrent or two DC LCB opening without command events occurr THEN TCMS requires roof line disconnectors opening to PCUs. After reactivation of the line voltage system the same pantograph shall be used. (50% of Propulsion)..	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR17	Second attempt of reclosing LCB (roof disconnector open): If AC line overcurrent is detected or a third DC LCB opening without command occurs within 5 seconds after re-closure of the LCB with roof disconnector open, the highvoltage system in the active half train shall be blocked TCMS requires the pantograph change to the driver (50% of Propulsion t).. If the LCB is not re opend on second attempt there is no further action needed.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR257	Third attempt of reclosing LCB (roof disconnector open and pantograph change): If a forth LCB re-opening within 5 seconds after closure no other action is possible, the high voltage system in the train is blocked. On the other hand if the LCB remain closed there is no further action needed (50% of propulsion)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR263	When a LCB opening without command occurs an event is created	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.363	If an AC overcurrent is detected an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR207	An EVENT shall be set when a short circuit (2 AC line overcurrent) is detected in AC configuration	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.728	The maximum current drawn by the vehicle in line voltage system shall be limited according the value set by the driver on IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_04.01.Zefiro-Europe.TRS.773
Description	The maximum allowed power to be extracted from the catenary by the train (also in multiple) shall be possible to be set by the driver at the IDU.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1050	The driver shall be able to limit the train current by enabling a current limiter device on the IDU: a) in 25 kV AC line voltage systems by steps of 20 A (in range of 20 to 560 A) b) in DC 3 kV line voltage systems by steps of 100 A (in range of 100 to 3000 A) c) in DC 1.5 kV line voltage systems by steps of 100 A (in range of 100 to 3200 A) In case of 15 kV line voltage system: 700 A In case of double traction the limits are doubled.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

2F_04.01.Zefiro-Europe.CONCEPT.HR344	TCMS shall send to PCUs not cutout the current limitation that each PCU has to respect	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1051	Current shall be limited according to given values from TCMS.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1152	TCMS shall set maximum level of current limitation after powered on: a) in 25 kV AC line voltage systems: 560 A b) in DC 3 kV line voltage system: 3000 A c) in DC 1.5 kV line voltage system: 3200 A In case of 15 kV line voltage system: 700 A In case of double traction the limits are doubled. Every time the catenary change, the max value for that catenary must be setted.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.3.3 High Voltage System Protection

3.3.3.1 *Function Design*

3.3.3.1.1 Normal Condition

HVSP supervises interference currents (e.g. 50Hz supervision in Italian DC line) and handles high voltage protections by voltage and current measures.

3.3.3.1.2 Failure Condition

1) Failure to detect line voltage.
2) Failure to measure overcurrent.
3) Other failure of HVSP.

3.3.3.1.3 Failure Supervision

1) Changeover to other HV system. Supervision by TCS.
2) Changeover to other HV system. Supervision by TCS.
3) HVSP is bypassed, supervision by TCS.

3.3.3.2 *Requirements*

ID	2F_04.01.Zefiro-Europe.TRS.309
Description	One line interference monitor per half-train shall be installed. (e.g. 50 Hz supervision in Italian DC line).
Safety level	SIL 2

Nr	**Description**	**Applicability Details**	**Derived to**	**User Interface**	**SIL**
2F_04.01.Zefiro-Europe.CONCEPT.112	At least one line interference monitor per half-train shall be installed for the supervision of the harmonic content for the DC high voltage line.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR285	In 3 kV configuration, If HVSP detects a 50 Hz Harmoni line current content > 1A for 3 seconds the line trip shall be open	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR286	In 3 kV configuration, If HVSP detects a 50 Hz Harmoni line current content > 1A for 3 seconds an event is created	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_04.01.Zefiro-Europe.TRS.691
Description	The HVSP shall realize the following functions: – Line current measurement – Return current measurement – Line voltage measurement – Self test
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.231	The HVSP shall realize the following functions: – Line current measurement AC – Return current measurement – Line voltage measurement – Self test	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR288	The HVSP latches the protection signals setting to TCMS for at least 800ms until a maximum of 2 seconds	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR311	If HVSP a supervision function disabling is received the function is disabled and its feedback is sent to TCMS	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR328	HVSP is allowed to re-close the line trip 2 seconds after the previous opening, unless the perevious opening has been due to AC line over current (ref. 2F_04.01.Zefiro-Europe.CONCEPT.225) or transformer differential overcurrent (ref. 2F_04.01.Zefiro-Europe.CONCEPT.1129).	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR241	In the maintenance and Driver menu shall be available a soft button to require the HVSP selftest. The button shall be active only if the LCB is open and all the pantographs lowered. If the button is pushed when the LCB is closed and/or any pantograph is raised a warning to operator is given	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.989	The HVSP Self test shall be requested if any of the following conditions are fulfilled: – Train has started up (after closing of Battery Circuit). – The HVSP is re-included after a bypass and the train is at standstill. – The self test is requested by the dedicated button on the TOD	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR173	10 hours before the selftest timeout HVSP shall send to TCMS a first warning regarding the remaining time to perform the selftest	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR232	An event shall be created when a first HVSP self test warning occurs	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR233	5 hours before the selftest timeout HVSP shall send to TCMS a second warning regarding the remaining time to perform the selftest	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR234	An event shall be created when a second HVSP self test warning occurs	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR236	The self test timeout shall be set 500**hours** after last successful test	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR242	An event shall be created when the HVSP self test timeout occurs	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR231	When the selftest is requested to HVSP the LCB shall be ordered to open (if closed) and	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	all pantographes shall be lowered (if raised) in the same 8-car train				
2F_04.01.Zefiro-Europe.CONCEPT.HR175	HVSP performs the selftest when the request is received from TCMS AND it is not bypassed AND the line trip is not energized (LCB Hold feedback high) AND the DC line voltage is lower than (<450V) and AC line voltage is lower than (<2kV)	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1131	As long as the HVSP self test in ongoing the LCB shall not be enabled to close and the pantograph shall not be raised. If the driver request to raise the pantograph while the HVSP self test is ongoing an event shall be set "HVSP self test ongoing" to inform the driver that the pantograph will be raised when the test is finished. The event shall be reset when HVSP self test is finished or the driver request to raise the pantograph is removed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1132	An event shall be set "HVSP self test finished" when the HVSP self test is finished to inform the driver that it is possible to raise the pantograph and to close the LCB.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1087	In case of HVSP self test error, HVSP will be bypassed (100% traction effort remaining in DC, 50% traction effort remaining in AC).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR202	In case of HVSP faulty, HVSP will be bypassed (100% traction effort remaining in DC, 50% traction effort remaining in AC).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR178	In case of HVSP faulty TCMS creates a HVSP failure event	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR179	In case of HVSP self test error involves only AC functionalities TCMS actives HVSP bypass only in AC configuration (50% traction effort remaining in AC).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR180	In case of HVSP self test error involves only DC functionalities TCMS actives HVSP bypass only in DC configuration(100% traction effort remaining).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR177	In case of HVSP life signal check fails from both ethernet lines the HVSP shall be bypassed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR341	In case of HVSP life signal check fails from both ethernet lines TCMS creates a HVSP communication failure event	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR176	if HVSP doesn't receive the selftest request command within 10 minutes of timeout HVSP shall set the failure status and keeps open the line trip	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.239	In case of HVSP self test error or HVSP is in faulty status: - If the failure involves only DC functionalities HVSP shall set the generic failure status AND the DC failure status AND not the AC failure status AND keeps open the line trip in DC configuration - If the failure involves only AC functionalities HVSP shall set the generic failure status AND the AC failure status AND not the DC failure status AND keeps open the line trip in DC configuration - If the failure involves all functionalities HVSP shall set the	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2

	generic failure status AND not the AC failure status AND not the DC failure status AND keeps open the line trip in any configuration -				
2F_04.01.Zefiro-Europe.CONCEPT.HR235	In case of HVSP is bypassed the pantograph is not allowed to raise in the same half train. In AC configuration the same half train has to be isolated	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR237	In case of HVSP is bypassed and the driver command to lift the pantograph in the same half train a message to perform the HVSP selftest is notified	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR174	In case of HVSP is bypassed an event to driver is notified to change the pantograph	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR243	When the HVSP is bypassed the bypass status is sent to HVSP	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.756	If an self test error is indicated, an EVENT shall be set.	Derived Requirement	4S_04.-.Power Supply	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR347	In case of HVSP is bypassed, if a Self test is requested by TOD, TCMS has to remove for 2 minutes the bypass of HVSP in order to run the self test if it is permitted.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

ID	2F_04.01.Zefiro-Europe.TRS.692
Description	The line circuit breaker shall open if – limits for line current exceeded (limits depending on country of operation) – Main transformer differential current exceeded – Short circuit detected – maximum line voltage exceeded
Safety level	SIL 2

Nr	**Description**	**Applicability Details**	**Derived to**	**User Interface**	**SIL**
2F_04.01.Zefiro-Europe.CONCEPT.234	The line trip shall open if – limits for line overcurrents in AC exceeded – Main transformer differential current exceeded – Short circuit detected – maximum line voltage exceeded – undervoltage line – 50 Hz harmonic current content in DC 3kV – self test error	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2

3.3.4 HVSP 50Hz protection bypass

3.3.4.1 *Function Design*

In case of detection of 50Hz current harmonic when the train is running on a DC line, the HVSP protection system opens the line circuit breaker to avoid potential risks.
Most probably this is not due to a failure of the train propulsion system but to some incoming noise directly from the power line.
The Italian regulation states that is possible to bypass this protection allowing the train to move to a different segment of line where this problem is not present.
This is allowed only when the train is running on a DC powered line.
A sealed switch is installed in the driver's cab to let the driver perform this action. The activation of this switch is managed by TCMS safe layer that communicates directly to the HVSP system to bypass this protection.

3.3.4.1.1 Normal Condition

In normal condition the switch is sealed and the 50Hz supervision functionality is not bypassed.

3.3.4.1.2 Failure Condition

1) The switch is turned but no bypass has been enabled.
2) The switch is not turned and the bypass has been enabled.

3.3.4.1.3 Failure Supervision

1) By the driver, if the protection occurs with the switch turned and the LCB cannot be closed.
2) By the driver, if the IDU indication is on without switch turned.

3.3.4.2 *Requirements*

ID	2F_04.01.Zefiro-Europe.TRS.792
Description	A sealed switch to bypass the HVSP 50Hz harmonic control (on DC lines only) shall be installed in the driver's cab.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1146	A sealed switch to bypass the HVSP 50Hz harmonic control (on DC lines only) shall be installed in the driver's cab.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1147	The HVSP bypass switch shall be connected to safe and redundant Digital Inputs.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1148	When the sealed switch is turned TCMS shall send the "Command: Disable HVSP 50Hz supervision" to HVSP via Ethernet bus.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1149	When the HVSP 50Hz supervision bypass is activated an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1150	When the HVSP 50Hz supervision bypass is activated an indication shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1151	The 50Hz supervision of HVSP (on DC line only) shall be bypassed when commanded by TCMS.	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR264	When the 50Hz supervision of HVSP bypass is selected the bypass has to be activated on both vehcles in a multiple composition.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

3.3.5 Neutral Section Passing

3.3.5.1 *Function Design*

3.3.5.1.1 Normal Condition

Neutral sections are powerless parts of the catenary system. There are two different types of neutral sections which are defined according to EN50388:

a) AC phase separation sections, which separate between two different phases of AC line voltage system.
   In Italy, crossing AC phase separation sections will be performed automatically by indications from ATP.
b) system separation sections, which separate between different line voltage systems (AC 25kV, AC 15kV, DC 3kV, DC 1.5kV)
   In Italy, crossing system separation sections will be initiated by ATP but driver has to select line voltage manually.

3.3.5.1.2 Failure Condition

1) AC phase separation section is not detected (ATP is faulty/isolated).
2) System separation section is not detected (ATP is faulty/isolated).
3) AC phase separation section detected but LCBs do not open.
4) System separation section detected but LCBs do not open.

5) System separation section detected but pantograph does not lower

3.3.5.1.3 Failure Supervisiong

1) Driver has to open LCB manually (Open LCB switch) before entering neutral section.
2) Driver has to open LCB manually (Open LCB switch) and lower the pantograph before entering neutral section.
3) Driver has to open LCB manually (Open LCB switch) before entering neutral section. When LCB is faulty, panto will lower immediately.
4) Driver has to open LCB manually (Open LCB switch) before entering neutral section.
   When LCB is faulty, panto will lower immediately.
5) Driver has to lower pantograph manually (Pantograph switch to "down position") after LCB opened but before entering neutral section.

3.3.5.2 *Requirements*

ID	2F_04.01.Zefiro-Europe.TRS.463
Description	The AC phase separation sections and separation section sequences shall be activated by External sensors e.g. ATP or track side magnets.
Safety level	SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.464
Description	When the AC phase separation section sequence is activated, the Line Converters shall be ramped down and the line circuit breakers shall be opened for the time of passage of the AC phase separation sections.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.515	AC phase separation section sequence shall be activated by external sensors (e.g. ATP).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1112	System separation section sequence shall be activated by external sensors (e.g. ATP).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1122	In italy, the sequence of crossing AC phase separation sections shall start when external sensors orders "traction block" and "LCB opening".	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1123	In Italy, the sequence of crossing system separation sections shall start when external sensors orders "traction block", "LCB opening" and "pantograph down".	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.521	For country options, the sequence of crossing AC phase separation sections or system separation sections shall start when the train has travelled the distance between the sensor in the leading unit and the raised pantograph.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.510	The power reference shall be ramped down to 0 before requesting the LCB to open prior to enter AC phase separation section or system separation section.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.441	The LCB shall be requested to be opened before entering the AC phase separation section or system separation section.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR265	When the Neutral or Separation command is received a backup braking is activated to keep the Auxiliary loads and the filter charged when the LCB is open, the converter first stage is switched off and the separation contactor is opened	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR284	If HVSP Master receives the Neutral or Separation section signal high THEN the line voltage supervision is disabled until the Line voltage selected is not changed.	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL0

	In particular the mastership is kept, the overvoltage and undervoltage protection are disabled, the signals “ACLineVOk” (if the selected syste is AC) or “DCLineVOk” (if the selected syste is DC) are kept				
2F_04.01.Zefiro-Europe.CONCEPT.HR20	The panto shall be lowered after requesting the LCB to open prior to enter system separation section.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.442	When the LCB has opened for AC phase separation section or system separation section it shall stay open for at least 4 seconds.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR21	When the LCB is opened entering AC phase separation section the Auxuliary Converter has to be supplied using the kinetic energy from backup braking activation	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR22	When the LCB is opened entering system separation section the Auxuliary Converter has to be supplied using the kinetic energy from backup braking activation	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.512	AC phase separation section or system separation section sequence shall be reset if the train is standing still.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.513	The AC phase separation section or system separation section sequence shall be reset on reset of setting condition	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.344	If LCB was opened on order from external sensors (ATP) before entering the AC phase separation section, the LCB shall be requested to be closed again after passing the AC phase separation section, automatically when ordered by external sensors only if any command to open hasn't been given from driver or internal protection of HV equipment and propulsion system, and system detection has confirmed selected line voltage system (see chapter 3.4.1).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.776	If LCB was opened on order from external sensors (ATP) before entering the system separation section, the LCB shall be requested to be closed again after passing the system separation section, automatically when ordered by external sensors (neutral signal from TCMS is reset) only if any command to open hasn't been given from driver (Enable LCB closing signal from TCMS is set) or internal protection of HV equipment and propulsion system, and system detection has confirmed selected line voltage system (see chapter 3.4.1).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR27	When the train arrives to the system separation section from AC Voltage line the PCU HV Master has to change the line trip configuration from AC to DC when receives that DC line voltage has been selected from TCMS	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR28	When the train arrives to the system separation section from DC Voltage line the PCU HV Master has to change the line trip configuration from DC to AC when receives that AC line voltage has been selected from TCMS	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR30	when the train is coming out from separation section entering in DC Line, after the LCB closing, the separation contactor shall be closed when the input filter of DC converter are charged or the precharge resistance shall be inserted to charge the input filter	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR31	when the train is coming out from separation system section entering in AC Line or from separation phase section, after LCB closing,	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0

	the separation contactor shall be closed if dc link is kept charged or the percharge resistance shall be inserted				
2F_04.01.Zefiro-Europe.CONCEPT.443	After the LCB has been closed automatically when ordered by external sensors (ATP), MCM power reference shall be ramped up.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.196	If the train is in single or multiple operation under 3 kV and the ATP is communication a voltage change (the catenary that will come is different than 3 kV) and if the voltage is going to 0 and the panto of the leading train is up. TCMS must command a fast panto down to all the panto in the composition. This to avoid that the train in sigle or multiple composition will run with the 3 kV panto raised under 25 kV.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR348	If ATP is not available (i.e. life on Ethernet KO or CEA is in excluded position), the neutral section crossing and HV system reconfiguration have to be performed manually by the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR349	If ATP is not available during neutral section crossing, an event shall be raised on TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

Passing a system separation section will be done in the same way as a AC phase separation section with the extra function to swap pantograph before reconnection on a different line voltage system.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.505	If system separation section is activated, from external sensors (ATP), LCB shall be ordered to open and pantograph shall be ordered to lower.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

The only action which is needed from driver during crossing of system separation section is the manual selection of the upcoming line voltage system (line voltage selector switch).

ATP will indicate to the driver which line voltage system will follow after the separation section.

ID	2F_04.01.Zefiro-Europe.TRS.708
Description	The AC phase separation or separation section shall be activated via External sensors (e.g. ATP). The External sensors also gives additional information to lower pantograph after the Line circuit breaker has been opened.
Safety level	SIL 0

The Italian ATP system reports: Information about the line voltage system on the other side of the neutral section.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.458	If external sensors (ATP) indicate a AC phase separation section requires lowering of pantographs, all pantographs shall be lowered before entering the AC phase separation section. (Note this is not the same as a system separation section)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.573	If external sensors (ATP) indicate that the voltage system after passing the AC phase / system separation section is different than the voltage system before the AC phase / system separation section, the passage shall be done according to requirement for system separation section.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.335
Description	Where AC phase separation sections require pantographs to be lowered and subsequently raised, these additional actions are permitted to be automatically initiated by the ATP.

Safety level	SIL 0

External sensors (ATP) might in some situation be able to not only order pantograph down but also pantograph up.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.523	If external sensors (ATP) require the pantograph to be raised, the selected pantograph shall raise. (This shall work in direct parallel to the "raise pantograph switch")	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1089	If the pantographs have been lowered on order from external sensor before entering a AC phase / system separation section and driver has not moved panto-switch to "down" position, raise-order for pantograph shall be blocked as long as the train has not passed neutral section.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1126	Pantograph shall only be allowed to raise, when TCMS has confirmed that selected line voltage system is in line with system which is indicated by ATP.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1127	It shall be indicated on IDU when selection of line voltage is not in line with ATP notification.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.462	If the pantographs have been lowered on order from external sensor before entering a AC phase / system separation section and driver has not moved panto-switch to "down" position, raise-order for pantograph after crossing the neutral section shall be set automatically by external sensors depending on selected line voltage system.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.687
Description	When multiple pantographs are lifted and spacing allowed in coming neutral section is exceeded, the pantographs shall be lowered before entering the neutral section.
Safety level	SIL 0

If two or more pantographs are lifted, a neutral section with a length about the same as the distance between two pantographs can be short circuit by the carbon strip connecting the neutral catenary with the live catenary.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.518	In multiple operation, pantographs shall raise when external sensors order “pantograph raising”.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.3.6 System Selection

3.3.6.1 *Function Design*

3.3.6.1.1 Normal Condition

The driver selects the line voltage system (AC 25kV, AC 15kV, DC 3kV or DC1.5kV) by using 4-position switch "Line voltage selector".
When line voltage system changes from one to another of these four voltage systems a system selection has to be performed. To identify exactly the country of operation, the country code needs to be selected.
The system detection units have to detect, to which line voltage system the train is connected.

3.3.6.1.2 Failure Condition

1) System detected by AC system detection unit and HVSP differ
2) Faulty System detected by AC system detection unit

3) System detected by DC system detection unit and HVSP differ
4) Faulty System detected by DC system detection unit
5) Plausibility fault DC and 3kV from HVSP

3.3.6.1.3 Failure Supervision

1) Faulty system detection. Lower and isolate AC pantograph. Driver has to raise other AC pantograph,
2) Faulty system detection. Lower and isolate AC pantograph. Driver has to raise other AC pantograph,
3) Faulty system detection. Lower and isolate DC pantograph. Driver has to raise other DC pantograph,
4) Faulty system detection. Lower and isolate DC pantograph. Driver has to raise other DC pantograph,
5) Faulty signals from HVSP. Lower and isolate DC pantograph. Driver has to raise other DC pantograph,

3.3.6.2 *Requirements*

ID	2F_04.01.Zefiro-Europe.TRS.771
Description	The following power supply system shall be possible to be selected by the driver at driver's desk: AC 25 kV 50 Hz, AC 15 kV 16.7 Hz, 3 kV DC, 1.5 kV DC.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.492	The driver shall be able to select the voltage system (15kV AC, 25kV AC, 3kV DC or 1.5kV DC) prior to raising the pantograph.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.976	A 4-position switch "Line voltage selector" shall be installed at the driver's desk. With this switch it shall be possible to select the following power supply systems: AC 25 kV 50 Hz, AC 15 kV 16.7 Hz, 3 kV DC, 1.5 kV DC.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_04.01.Zefiro-Europe.CONCEPT.977	The following positions of the "Line voltage selector" switch shall be connected to TCMS AC 25 kV 50 Hz, AC 15 kV 16.7 Hz, 3 kV DC, 1.5 kV DC.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1069	The voltage selection has to be stable for 1 second before TCMS uses the input signal	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL2
2F_04.01.Zefiro-Europe.CONCEPT.HR23	TCMS shall send the line voltage selected to PCUs.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR24	TCMS shall send the line voltage selected to HVSP.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

In addition country code is needed to identify exactly, in which country the train operates.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.731	If the train is not designed for operation in all 4 line voltage systems and the driver turns "line voltage selector" to a system which is not implemented, the raise of pantographs shall be blocked.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.732	If the driver turns "line voltage selector" to change the selected line voltage system without opening the LCB and lowering the pantograph before, all LCB shall be requested to open and all pantographs shall be lowered immediately.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.700
Description	It shall be ensured that the DC measurement device is not damaged if a DC panto is raised at an AC catenary by mistake.
Safety level	SIL 1

There will be a system detection device at each TT2, T4, T5 and TT7 car. Each will be able to detect as well AC and DC line voltage regardless of what pantograph it is connected to.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.553	System detection device shall detect and report both AC and DC voltage system.	Derived Requirement	4S_04.-.Power Supply		SIL 2

ID	2F_04.01.Zefiro-Europe.TRS.337
Description	When pantographs are not lowered from the catenary line when passing a neutral section, only those electric circuits on the traction units, which instantaneously conform to the power supply system at the pantograph, may remain connected.
Safety level	SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.707
Description	The high voltage system system shall protect itself against false voltage system selection.
Safety level	SIL 0

When the LCBs and system isolation switches are open, the only equipment that is connected to the line voltage via the pantograph is the system detection device. This device detects if the present line voltage system (AC 25kV, AC 15kV, DC 3kV, DC 1.5kV) is in line with the driver-selected line voltage system.
If present line voltage system does not comply to selected system, pantograph will be lowered and driver gets an indication to select another line voltage system.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.739	In case the detected system does not fit the selected system – there shall be a message on the IDU – the raise signal shall be canceled and the pantograph become lowered	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.524	If line voltage system is AC and 50Hz is detected propulsion system shall be set up for 25kV 50Hz.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.525	If line voltage system is AC and 16 2/3Hz is detected propulsion system shall not be set up for 15kV 16 2/3Hz and the pantograph become lowered	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.526	If line voltage system detected is DC and line voltage selected is 3kV propulsion system shall be set up for 3kV DC.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.527	If line voltage system detected is DC and line voltage selected is 1.5kV propulsion system shall be set up for 1.5kV DC.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR25	PCUs shall send the propulsion configuration depending on voltage selected to TCMS	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.287
Description	For AC and for DC it shall be detected which line voltage system is connected.
Safety level	N/A

System selection is done automatically by comparing the measured voltage system with the driver-selected voltage system. The system validates the selected voltage system via measurement before allowing the LCBs to be closed.
The system detection detects the voltage of contact wire voltage (15kV AC, 25kV AC, 3kV DC or 1.5kV DC).

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.219	The LCB shall be opened resceptively not closed if the system detection detects another line voltage system (15kV AC, 25kV AC, 3kV DC or 1.5kV DC) than selected.	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR26	The LCB shall be opened resceptively not closed if the system detection detects another line voltage system (15kV AC, 25kV AC, 3kV DC or 1.5kV DC) than selected.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.312	An event shall be set if the line voltage system detected by HVSP differs from the selected line voltage system by the line voltage selector.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_04.01.Zefiro-Europe.TRS.702
Description	It shall be ensured that the main trafo is not damaged if a AC panto is raised at a DC catenary by mistake.
Safety level	SIL 1

The system detection device will prevent the LCB from closing if the measured system is not the valid voltage system for that LCB.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.459	It shall not be possible to close the LCB when wrong voltage is detected compared to the raised pantograph.	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR33	It shall not be possible to close the LCB when wrong system is detected compared to the raised pantograph.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.785	If line voltage system sent from Propulsion (HVSP OR PCU) does not comply to selected system by the line voltage selector the closing of LCB shall be disabled and the raised pantograph lowered.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.4 Line Circuit Breakers and system isolation switches

3.4.1 Line Circuit Breaker Closing

3.4.1.1 *Function Design*

3.4.1.1.1 Normal Condition

Connecting of line voltage by closing LCBs.

3.4.1.1.2 Failure Condition

1) LCB does not close on order.
2) LCB closes without order.

3.4.1.1.3 Failure Supervision

1) If the feedback "LCB closed" is not given latest 1s after the "LCB close" order is set, the LCB is considered as faulty. The LCB will be blocked but "activation of line voltage" will retry the closing order.
   After 3 attempts without success within 30 minutes the LCB will be blocked permanently and the pantograph will be blocked.
2) The related pantograph will be lowered, the pantograph disconnector will open and both components will be blocked!

3.4.1.2 *Requirements*

ID	2F_04.01.Zefiro-Europe.TRS.468

Description	Line circuit breaker shall be closed when – 'Line circuit breaker close' command given by driver OR automatically when leaving a neutral section AND – pantograph is already lifted AND – no interlocking for maintenance activities active AND – no line circuit breaker opening command The 'Line circuit breaker close' command is done by turning the Line Circuit Breaker switch on driver's desk to position ON.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.978	There shall be 3-position switch for closing/opening of the Line Circuit Breaker. The positions shall be: 1: LCB close (unstable) 2: neutral (stable) 3: LCB open (unstable)	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_04.01.Zefiro-Europe.CONCEPT.979	The following positions of the Line Circuit Breaker switch shall be connected to TCMS: 1: LCB close 2: LCB open	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.127	The status of the LCBs shall be displayed on the IDU: – open/closed – faulty (automatic block) – manually blocked	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.970	The LCB shall be inhibited to close if there is no valid country code (country code = 0) or not available country code.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.374	When the driver turns LCB-switch to "close" position AND the pantograph is raised AND line voltage is detected for the selected line voltage system (AC 25kV, AC 15kV, DC 3kV or DC 1.5kV), the LCB which is related to the raised pantograph shall be enabled to close. LCB can be closed from driver's desk only if the master controller is on zero position.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1224	If the LCB is requested (from driver) to close while the Master controller is out of the 0 position, an advisory shall be shown to the driver asking 'to set the master controller in 0 position before ordering LCB closure'	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1090	AC LCB shall only be allowed to close when both AC roof line disconnectors are in end position. In case of plausibility fault of remote AC roof line disconnector, both roof disconnectors have to be oredered open with LCB open.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1091	DC LCB shall only be allowed to close when both DC bus bar disconnectors are in end position. In case of plausibility fault of DC roof line disconnector, both roof disconnectors have to be ordered open with LCB open.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.992	The LCB shall only be allowed to be closed when the LCB enable signal is set from TCMS.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR244	If both roof disconnector are closed in AC or DC before LCB closing the line trip has to be configred in global mode in both Half Train	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A

2F_04.01.Zefiro-Europe.CONCEPT.HR245	If both roof disconnector are open in AC or DC before LCB closing the line trip has to be configred in local mode in both Half Train	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR246	Before LCB closing the HV Master PCU has to set the line trip in Master Mode	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR247	In DC Voltage configuration, before LCB closing the HV Master PCU has to set the LCB for the right Voltage system: 1.5kVDC or 3kV DC	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR249	DC LCB configuration has to be changed by a 1 second pulse command only if the current status is not coherent with the line Voltage selected	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR250	DC LCB needs a peak of current to close given setting for 1 the LCB DC close command	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR251	The filters precharge sequence starts after 1 second after LCB closing	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR252	Each PCU filter precharge is delayed by 500ms from the previous	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR317	The Converter equipped with ACM module starts the filter precharge before the others	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A

There are situations when the LCB will remain open even though TCMS is enabling closing of the LCB. The decision to close the LCB will based on internal Propulsion supervision and the enable order from TCMS.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.500	The system isolation switch shall be closed before the DC LCB can be closed.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.381	When ever the “LCB Close” order is removed, there shall be a 4 second delay before the LCB is allowed to close again.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.386	The number of closings of each LCB shall be stored as condition data.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.532	When the “LCB closed” feedback and “LCB not closed” feedback indicates different states (one indicates closed and one open) for more than 1 second the LCB shall be ordered to open.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.387	When the “LCB closed” feedback and “LCB not closed” feedback indicates different states (one indicates closed and one open) for more than 1 second an event shall be set.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.531	When the “LCB closed” feedback and “LCB not closed” feedback indicates different states (one indicates closed and one open) for more than 1 second IDU shall show a faulty LCB.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.761	When the feedback "LCB closed" is not given latest 1 second after the "LCB close" order is set, the LCB is considered as faulty. The LCB shall be blocked, but operation	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A

	of "Close LCB" switch shall retry the closing order. After 3 attempts without success within 30 minutes the LCB will be blocked permanently and the pantograph will be blocked.				
2F_04.01.Zefiro-Europe.CONCEPT.762	If the LCB is closed without order, the related pantograph shall be lowered, the pantograph disconnector shall open and both components shall be blocked!	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1161	After manual **exit from parking mode**, LCB under selected pantograph has to be closed and LCB under not selected pantograph has to be opened. – In case the new configuration is different from the LCB closed during parking mode and **DC line voltage** system is selected, the LCB under selected pantograph shall be automatically closed and afterwards the LCB under not selected pantograph shall open. In case in the new configuration the driver has requested raising of both pantographs, this procedure lets closed the LCB which is related to the rear pantograph. – In case the new configuration is different from the LCB closed during parking mode and **AC line voltage** system is selected, LCB closing command will not be given automatically by TCMS (Driver has to do it manually) This procedure can be only performed at standstill. If the train starts running, this procedure will be stopped immediately opening both LCBs.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1162	While manual exit from parking mode procedure is active, LCB shall close on request from TCMS (EnableLCBClosing), if all the other conditions are fullfilled, and it allowed to close both LCBs for a short time only (TBD) if DC line is selected. The LCB which is no longer requested to close from TCMS, shall open without any interruption of the auxiliary power supply.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A

2F_04.01.Zefiro-Europe.CONCEPT.1164	The voltage coherence of the two pantographs shall be confirmed, before to allow both LCB to be closed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR29	HVSP Master close also the second line trip to permit the second LCB closing if Enable2LCBClosing is received from TCMS AND the voltage at the pantograph is coherent with the system selected (only under DC voltage)	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR59	if Enable2LCBClosing is received from TCMS AND the voltage at the second pantograph is coherent with the system selected (DC), PCU Half Train Master close closes the linetrip bypass and when closed sets the ReadyClose2Lcb	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR342	When TCMS receives the ReadyClose2Lcb signal from the Half Train Master PCU sends the LCBCloseEnable on that half train to command the second LCB closing	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR59	When the second LCB is closed, TCMS reset the LCB enable closing sent to the current HV Master PCU and reset the pantograph up command on the same Half Train	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR190	PCU Half Train Master configures the line trip in Master mode and becomes HV Master before opening the line trip by pass	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR191	PCU HV Master with LCB open configures the line trip in Slave mode and resets its HV master status	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR266	HVSP open the second line trip when Enable2LCBClosing signal received from TCMS is reset	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.388	In operation mode (refer to key interlocking system), closing of LCB shall be only possible if pantograph is raised and selected line voltage system is confirmed by system detection.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.495	In maintenance mode (refer to key interlocking system), closing of LCB shall be possible if ALL pantographs are lowered.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

2F_04.01.Zefiro-Europe.CONCEPT.1177	An event shall be set in case the maintenance mode relays have discrepancy between DM1 and DM8 on the same trainset	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1059	In maintenance mode (refer to key interlocking system), closing of LCB shall be possible if ALL pantographs are lowered.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.796	LCB shall only be allowed to close without charging resistor inserted when the input filter of DC converter is charged or in AC when the DC-link is charged to avoid hard switch-on of converters.	Derived Requirement	4S_04.-.Power Supply (PropulsionControl)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.389	When external power supply is active the LCB shall not be allowed to close.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1214	If the driver,with train standtill and with external power supply,, request any panto to be rised a warning to the driver shall be set. The warning alerts the driver that at least one 400V external plug is inserted and in case of false reporting the plug can be manually isolated using softkey on iDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.703
Description	Line circuit breaker shall only be allowed to close if valid line voltage is detected
Safety level	SIL 1

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.347	Line circuit breaker shall only be allowed to close if valid line voltage is detected or train is in maintenance mode. See section 3.4.1.2	Derived Requirement	4S_04.-.Power Supply		SIL 1
2F_04.01.Zefiro-Europe.CONCEPT.1163	During parking mode, in case of missing line voltage the automatic exit procedure starts. In case the line voltage is detected again, the automatic exit procedure shall be stopped and LCB shall be re-closed (only LCB already closed before missing line voltage, no other LCB shall be closed) To avoid simultaneous closing of many trains, LCB shall be automatically closed after a defined amount of time from the line voltage return that is proportional to the train number: time delay in seconds = (train number) + 5, where (train number) is integer in the range 0 to 99, considering the two least significant digits only.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.623
Description	The line circuit breakers of the train shall be closed one after another to limit inrush currents.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL

2F_04.01.Zefiro-Europe.CONCEPT.249	If there are 2 trains in multiple the "LCB Close" order shall be delayed 1 second in the second train.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.705
Description	The Line circuit breaker closure shall be synchronised to close at the top of the sinusodial line voltage of 15kV AC. This shall avoid over magnetisation of the main trafo.
Safety level	SIL 1

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.350	The Line circuit breaker closure shall be synchronised to close at the top of the sinusodial line voltage of 15kV AC. This shall avoid over magnetisation of the main trafo.	Derived Requirement	4S_04.-.Power Supply		N/A

3.4.2 Line Circuit Breaker Opening

3.4.2.1 *Function Design*

3.4.2.1.1 Normal Condition

Disconnection of Line Voltage will be done by LCB.

3.4.2.1.2 Failure Condition

1) LCB does not open on order.
2) LCB opens without order.

3.4.2.1.3 Failure Supervision

1) The related pantograph will be lowered, the pantograph disconnector will open and both components will be blocked!
2) If the LCB opens without order it will be ordered to close again. If the failure is present 3 times within 30 minutes the LCB will be blocked permanently and the pantograph will be blocked.

3.4.2.2 *Requirements*

ID	2F_04.01.Zefiro-Europe.TRS.234
Description	Line circuit breaker shall be opened when – open command from HVSP OR – line voltage system detection protection or – interlocking for maintenance activities is active OR – drivers emergency push button was pressed OR – ADD on pantograph or – protective shut down from Propulsion.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.408	If a line trip is detected the LCB shall be opened.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR316	The PCU sends the line trip relay status to TCMS as the status of the series of the included MOCUs line trip relay.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.750	If a power supply / propulsion line trip is detected the LCB shall be opened.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.409	If the “Emergency push button” is pushed the LCB shall be opened via Train Line (“LCB Emergency Off”).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

2F_04.01.Zefiro-Europe.CONCEPT.751	If the "Emergency push button" is pushed the LCB shall be opened via Train Line ("LCB Emergency Off").	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1179	If the "Emergency push button" is pushed and the LCB Emergency Off activation is not read in every line trip chain of whole composition an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1071	The LCB shall open immediately, when pantograph down order is given.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.410	The LCB shall open automatically, if any A-Key is activated (key interlocking system).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.752	The LCB shall open automatically, if any A-Key is activated (key interlocking system).	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1180	If any A-Key is activated and the LCB Emergency Off activation is not read in every line trip chain of whole composition an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1181	When IDU soft key is pressed from maintainer, a test of LCB Emergency Off trainline activation shall be carried out (energizing dedicated DOs only in the active cab)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1182	Automatically every time when the HVSP test is performed, a test of LCB Emergency Off trainline activation shall be carried out (energizing dedicated DOs only in one cab at the head of composition)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1183	If LCB Emergency Off trainline activation test is performing and the LCB Emergency Off activation is not read in every line trip chain of whole composition an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1092	When one AC roof line disconnector leaves end position, AC LCB shall be ordered to open.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1093	When one DC bus bar disconnector leaves end position an event shall be set when plausability fault is detected.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.763	When LCB does not open on order, the related pantograph shall be lowered, the pantograph disconnector shall open and both components shall be blocked!	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.764	When the LCB opens without order, it shall be ordered to close again. If the failure is present 3 times within 30 minutes the LCB will be blocked permanently and the pantograph shall be blocked.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A

ID	2F_04.01.Zefiro-Europe.TRS.704
Description	Line circuit breaker shall be opened when – 'Line circuit breaker open' command given by driver OR – entering neutral section
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.353	When the Line Circuit Breaker switch is turned to "LCB open" the LCB shall be ordered to open (also in rear train in case of multiple operation).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1056	LCB shall open when country code becomes invalid (country code = 0).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.403	A permanently blocked (due to a failure) LCB shall be blocked until a master reset is made or PCU is restarted.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A

<table>
<tr><td>2F_04.01.Zefiro-Europe.CONCEPT.995</td><td>If LCB is permanently blocked (due to failure), related pantograph shall be ordered to lower and related pantograph disconnector shall open and both, pantograph and disconnector, shall be blocked.</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 0</td></tr>
<tr><td>2F_04.01.Zefiro-Europe.CONCEPT.406</td><td>If the feedback ‘LCB Closed’ changes to ‘LCB not closed’ while the ‘LCB close' order is high, the status of the LCB is inconsistent and the LCB shall be ordered to open and the LCB shall be permanently blocked.</td><td>Derived Requirement</td><td>4S_04.-.Power Supply (Propulsion Control)</td><td></td><td>SIL 0</td></tr>
<tr><td>2F_04.01.Zefiro-Europe.CONCEPT.537</td><td>If the feedback ‘LCB Closed’ changes to ‘LCB not closed’ while the ‘LCB Close' order is high, the status of the LCB is inconsistent and an event shall be set.</td><td>Derived Requirement</td><td>4S_04.-.Power Supply (Propulsion Control)</td><td>EVENT</td><td>N/A</td></tr>
<tr><td>2F_04.01.Zefiro-Europe.CONCEPT.986</td><td>If all LCBs in the trains are open (valid also for Multiple Unit), the 'LCB is open' indication lamp shall be illuminated on driver's desk.

For more details see the tables below for multiple unit and single unit:

single unit:
<table>
<tr><td colspan="3">SINGOLO</td></tr>
<tr><td>IR1</td><td>IR2</td><td>spia</td></tr>
<tr><td>0</td><td>0</td><td>accesa</td></tr>
<tr><td>0</td><td>1</td><td>spenta</td></tr>
<tr><td>1</td><td>0</td><td>spenta</td></tr>
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multiple unit:
<table>
<tr><td>IR1</td><td>IR2</td><td>IR3</td><td>IR4</td><td>Spia</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>Accesa</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>1</td><td>LAMP</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>LAMP</td></tr>
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1 = chiuso</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 0</td></tr>
<tr><td>2F_04.01.Zefiro-Europe.CONCEPT.985</td><td>When ordered by TCMS, the indication lamp 'LCB is open' shall be illuminated on the driver's desk.</td><td>Derived Requirement</td><td>4S_09.02.09.-.Conventional Train Control - Circuits</td><td>HW HMI</td><td>SIL 0</td></tr>
<tr><td>2F_04.01.Zefiro-Europe.CONCEPT.727</td><td>If the cab is deactivated without entering the parking mode, all LCB shall be reqested to open.</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 0</td></tr>
</table>

ID	2F_04.01.Zefiro-Europe.TRS.712
Description	Line circuit breaker shall be opened fast enough when system change section is entered under speed to avoid damage of electric. Equipment
Safety level	SIL 1

If the train is passing a system separation section with a manual indication in proper time or a correct signal from external sensors, the LCB will open before entering the system separation section. See section 3.3.4

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1105	IF HVSP opens the line trip due to undervoltage protection in DC AND the train speed is >30km/h AND ATP communication is in fault THEN the pantograph shall be lowered using the pantograph fast down trainline after the LCB has opened	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR345	IF TCMS resets the LCB Enable due to DC line voltage out of range because too low AND the train speed is >30km/h AND ATP communication is in fault THEN the pantograph shall be lowered using the pantograph fast down trainline after the LCB has opened	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1070	The pantograph fast down trainline shall be set when ordered by TCMS DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		N/A

ID	2F_04.01.Zefiro-Europe.TRS.714
Description	Line circuit breaker shall open in case of risk for overtemperature for trafo
Safety level	SIL 1

ID	2F_04.01.Zefiro-Europe.TRS.624
Description	In case of normal opening (no protection situation) the Line Converter Modules shall be ordered to ramp down before opening the line circuit breaker
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.252	Any order of normal opening of LCB requires a ramp down of line converters before LCB can be opened.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A

ID	2F_04.01.Zefiro-Europe.TRS.326
Description	In case of fire, the power supply to the affected system shall be interrupted. If one traction circuit is concerned, the respective line circuit breaker shall open, the affected system isolated from the traction circuit and the line circuit breaker then close again.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.255	If fire is detected in the underframe the LCB shall be requested to open and the disconnectors shall be requested to open (TCMS to PCU via bus).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.465	If the LCB and roof line disconnectors has opened due to fire, this half of the train where the fire is detected shall be isolated (disable closing of LCB and block raising of pantograph) and an event shall be set to inform the driver that change of pantograph is required.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1100	It shall be possible to manually cut out a LCB.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1101	If the LCB is manually cut out the LCB an event shall be set as long as the LCB is cut out.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1102	A LCB that is manually cut out shall be opened and not closed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

<table>
<tr>
<td>2F_04.01.Zefiro-Europe.CONCEPT.1245</td>
<td>
After 1 sec. that any AC LCB is open and not closed in the consist (single or multiple operation) and a voltage of AC is detected, an event shall be set and the related panto must be blocked. The line voltage to take in consideration shall be the one after the AC LCB that is provided by a PCU variable. The supervision shall be deactivated if AC LCB is closed and not open or after 10 seconds that the AC LCB is open and not closed.
<table>
<tr><th>PCU -> CCU-O</th><th>Signal</th><th>Cycle time</th><th>Description</th><th>Scale</th></tr>
<tr><td>CX1CT607</td><td>HCV_XU_AcLn</td><td>64 ms</td><td>LCM AC line voltage to remote system</td><td>1=10V</td></tr>
<tr><td>CX1CT6C2:7</td><td>HCV_VU_AcLn</td><td>64 ms</td><td>Valid bit: LCM AC line voltage to remote system</td><td>L</td></tr>
</table>
The function is activated if the voltage is higher than 15 kV for more than 0.5 sec (under 25 kv operation) after 1 sec. even if any LCB is in open position, the panto related to the LCB that was in closed position before must be put in fault.
An event to the driver shall be set with the following sentence” LCB X non aperto, il pantografo è inibito e non deve essere effettuato nessun master reset”. Also the buzzer shall be activated.
This logic must be done on consist level, this means that if an AC LCB on the rear/front train is not opened , the message must be send to the driver in the front train where the cab is active.
The massage for the diver are:
For the fault of the LCB in T4
"Isolare pneumaticamente l’impianto di sollevamento pantografo 25 kV tramite rubinetto nel quadro 59.C4 (rubinetto 08.18/02) della carrozza segnalata"
For the fault of the LCB in T5
"Isolare pneumaticamente l’impianto di sollevamento pantografo 25 kV tramite rubinetto nel quadro 59.C4 (rubinetto 08.18/02) della carrozza segnalata"
The eviroment data related to the event shall be the folloving, with a sample time of 100 ms if possible
<table>
<tr><td>Name</td><td>Description</td></tr>
<tr><td>DBC_CCV_W_TrBrEff</td><td>Traction brake effort request</td></tr>
<tr><td>DBC_BSV_XF_IduAchTrEffCst</td><td>Sum of achieved traction effort</td></tr>
<tr><td>DBC_BSV_XF_IduAchEdBrCst</td><td>Sum of achieved ED brake effort</td></tr>
<tr><td>DBC_BSV_XF_BcuBrAchCst</td><td>Sum of achieved DEP brake effort</td></tr>
<tr><td>DBC_CCT_X_SpdMdRef</td><td>Train speed reference</td></tr>
<tr><td>DBC_CCV_Xv_SpdLim</td><td>Speed limitation</td></tr>
<tr><td>MPW_BCV_S_LnTrp_T4</td><td>Status, line trip</td></tr>
<tr><td>MPW_BCV_S_LnTrpCmpl_T4</td><td>Status, line trip complete</td></tr>
<tr><td>MPW_BCV_C_TstMdLnTrpTst_T4</td><td>Test mode, line trip test</td></tr>
<tr><td>MPW_BCV_S_LnTrpOk_T4</td><td>Status, line trip test is OK</td></tr>
<tr><td>MPW_BCV_S_LimTstToUnt1</td><td>Status: Line trip test Time out</td></tr>
<tr><td>MPW_BCV_S_LimTstWaUnt1</td><td>Status: Line trip test Warning</td></tr>
<tr><td>TC_BI_1PCU1 _VU_LcmAcLnRmSys</td><td>Valid bit: LCM AC line voltage in T4 from PCU1</td></tr>
<tr><td>TC_BI_3PCU2 _VU_LcmAcLnRmSys</td><td>Valid bit: LCM AC line voltage in T4 from PCU3</td></tr>
<tr><td>TC_BI_1PCU1_S_HVMst</td><td>Status: Local PCU1 is High Voltage Master, Unit1</td></tr>
<tr><td>TC_BI_3PCU2_S_HVMst</td><td>Status: Local PCU3 is High Voltage Master, Unit1</td></tr>
<tr><td>TC_BI_1PCU1 _XU_LcmAcLnRmSys</td><td>LCM AC line voltage in T4 from PCU1</td></tr>
<tr><td>TC_BI_3PCU2 _XU_LcmAcLnRmSys</td><td>LCM AC line voltage in T4 from PCU3\n</td></tr>
<tr><td>MPW_BCV_S_Pcu1LcbOpd</td><td>LCB is opened from PCU1</td></tr>
<tr><td>MPW_BCV_S_Pcu1LcbCld</td><td>LCB is closed from PCU1</td></tr>
<tr><td>MPW_BCV_S_Pcu3LcbOpd</td><td>LCB is opened from PCU3</td></tr>
<tr><td>MPW_BCV_S_Pcu3LcbCld</td><td>LCB is closed from PCU3</td></tr>
</table>
</td>
<td>Derived Requirement</td>
<td>4S_09.03.05.-.TCMS Software</td>
<td></td>
<td>SIL 0</td>
</tr>
</table>

3.4.3 Isolate High Voltage

3.4.3.1 *Function Design*

3.4.3.1.1 Normal Condition

It is possible to isolate short circuits or failures to keep half of the traction of an 8 car train available.
The following table summarizes the roof disconnector handling when a failure condition occurs

Open Command	Closing command	Open Status Local	Open Status Remote	Closed Status Local	Closed Status Remote	Undefined Status Local	Undefined Status Remote	Action	Diagnostic
-	x	-		-	-	x	x	(3 tentativi di chiusura) ->Open both disconnector	Faulty
-	x	-	-	x	-	-	x	Open both disconnector	Faulty (Remote)
-	x	-	-	-	x	x	-	Open both disconnector	Faulty (Local)
x	-	-	-	-	-	x	x	LCB cannot closed (3 tentativi di apertura)	Faulty
x	-	x	-		-	-	x	Local LCB is allowed to close	Faulty (remote)
x	-	-	x	-	-	x	-	LCB is allowed to close only on the remote HT	Faulty (local)
x			x	x				LCB is allowed to close only on the remote HT	Faulty (local)
x	-	x	-	-	x	-	-	-	Faulty (remote)
-	x	x	-	x	-	-	-	Open both disconnector (dopo 3 tentativi di chiusura)	Local Not closing
-	x	-	x	x	-	-	-	Open both disconnector (dopo 3 tentativi)	Remote Not closing

3.4.3.1.2 Failure Condition

1) One disconnector does not open on order.
2) One disconnector does not close on order.
3) Position of disconnectors not known.

3.4.3.1.3 Failure Supervision

1) Roof line disconnector does not open on order. Event will be set and driver must change pantograph manually.
2) Both Roof line disconnector shall be open. Event will be set.
3) Roof line disconnector is in an unknown position. Event will be set and driver must change pantograph manually.

3.4.3.2 *Requirements*

ID	2F_04.01.Zefiro-Europe.TRS.474
Description	It shall be possible to isolate separate high voltage sections.
Safety level	SIL 0

A train half is considered "isolated from DC catenary" when the DC line disconnector related to active pantograph is open and either its DC pantographs are lowered or the system isolating switch is open.
A train half is considered "isolated from AC catenary" when the AC line disconnector related to active pantograph is open and either its AC pantographs are lowered or the AC LCB is open.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.997	The roof line disconnector shall be considered to **be not opening** on order if both feedbacks indicate closed position after: – 3 seconds for AC disconnector – 1 second for DC disconnector	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A

	Three attempts shall be allowed to open with 10 seconds interval between each attempt. If after 3 attempts the roof line disconnector has not opened on order the roof line disconnector shall be indicated **as faulty** to TCMS. **both disconnectors shall be ordered to open.**				
2F_04.01.Zefiro-Europe.CONCEPT.998	The roof line disconnector shall be considered to **be not closing** on order if both feedbacks indicating open position after: – 3 seconds for AC disconnector – 1 second for DC disconnector Three attempts shall be allowed to close with 10 seconds interval between each attempt. If after 3 attempts the disconnector has not closed on order then **both disconnectors shall be ordered to open.**	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.999	The roof line disconnector shall be considered to be in an **unknown position** if neither open or close acknowledge is set or open and close acknowledge are set simultaneously for 4 seconds. Three attempts shall be allowed to open with 10 seconds interval between each attempt. If the position is still unknown after 3 attempts roof line disconnectors shall be indicated **as faulty** to TCMS. **both disconnectors shall be ordered to open.**	Derived Requirement	4S_04.-.Power Supply (Populsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1222	If the system isolation switch is indicated as faulty from the PCU (propulsion control unit) then the pantograph related to the faulty switch shall be blocked and an event shall be set to indicate to the driver that pantograph change is required.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.474	The roof line disconnectors shall be closed at startup of train	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1223	The TCMS, since the start up of the train till the shut down, has to send the status of the battery contactor closed to the propulsion system.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.475	The status (open/closed) of the roof line disconnectors shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.137	If a failure in HV-part is detected, which requires separation of both halves of the train (e.g. earth fault), the roof line disconnector related to active pantograph shall be requested to open.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR350	The PCUs have to send the information of earth fault to TCMS, when the PCUs detected the earth fault.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR320	If the request to isolate the AC system is received from the PCU half train Master the local pantograph AC shall be not used (lowered if raised) and the AC roof disconnector opening request has to be sent to PCUs	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR321	If the request to isolate the DC system is received from the PCU half train Master the local pantograph DC shall be not used (lowered if raised) and the DC roof disconnector opening request has to be sent to PCUs	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.467	The DC-line disconnectors shall only be operated when all pantographs are lowered or DC LCBs are opened.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.586	The AC-line disconnector shall only be operated when all AC pantographs are lowered or all AC LCBs are in open position.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1226	the IDU has to provide a cut in/out virtual button for the driver to request the manual opening of the the roof disconnectors 3 or 25 kV. This	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

	button must be selective for train and line voltage.				
2F_04.01.Zefiro-Europe.CONCEPT.1227	When the TCMS recives the roof disconnector **cut out request from TOD**, the cut out request has to be sent to propulsion only if: – Train is standstill AND – All panto are down OR All LCB are open	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1228	When the roof disconnector cut out request is ordered only the selected voltage (TCMS has to check the driver desk selector) roof disconnector shall be opened. Note: if 3kv is selected and the driver request the roof disconnector opening only the DC disconnector request has to be sent to propulsion (see icd variable for request)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1231	When the roof disconnector opening request is ordered from the driver the TCMS implements the following actions: – a Master reset is sent to all equipments ? – 10 s after the master reset the TCMS request to propulsion to open both (3 kV or 25kV) roof disconnector	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1229	In case the roof disconnector opening order is triggered to propulsion an event shall be set for enginering	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1230	In case the the driver request a roof disconnector opening but not all conditions are true a selective (one event for each condition not verified) event shall be set to the driver	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1232	The roof disconnectors can be **cut in** if: – a cot out command has been previously delivered AND – Train is standstill AND – All panto are down AND – All LCB are open	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1233	In case the roof disconnector closing order is triggered to propulsion an event shall be set for enginering	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1234	In case the the driver request a roof disconnector closing but not all conditions are true a selective (one event for each condition not verified) event shall be set to the driver	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR253	AC roof disconnector has to be closed keeping high the closing command until the closed feedback is high with a 5 seconds timeout on the command.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR254	AC roof disconnector has to be opened keeping high the opening command until the opened feedback is high with a 5 seconds timeout on the command.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR255	DC roof disconnector has to be closed keeping high the command for 2 seconds	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR256	DC roof disconnector has to be opened keeping high the command for 2 seconds	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A

3.6 Main Transformer cooling and temperature supervision

The main transformer is cooled by running the transformer oil pump and cooling fans.
The transformer cooling shall be activated enabling start of transformer oil pump when the following conditions are fulfilled:
- The AC or DC LCB is closed AND
- The 3-phase bus is active AND
- Not too low oil level AND
- Not overpressure

OR (in DC o in mantenimento la pompa non è attivata, solo per temperatura? Attiverei sempre con 3-fase presente. Verificare sul 1000 attuale)
- Oil temperature is high [AND
- The 3-phase bus is active AND
- Not too low oil level AND
- Not overpressure

The transformer cooling shall be stopped without any time delay at too low oil level or at transformer overpressure
Main transformer supervision is redundant with two PCU's supervising the transformer in each half train.
A redundant propulsion dedicated RIO-TR unit is used to read the status of oil flow, oil level and oil pressure. The oil temperature sensors are acquired from TCMS and sent to PCUs. The RIO-TR unit is also used for fan and pump control and supervision
The temperature supervision is handled by TCMS and PCU. The TCMS open the LCB if a too high temperature value is reached, the PCU limits the traction/braking power when the temperature is high (value lower than the threshold used by TCMS to open the LCB)

3.6.1 Main Transformer cooling control

The main transformer is cooled by running the transformer oil pump and cooling fans The transformer oil cooling fan control manages two fan modules for half train at two different speed depending on the oil temperature and vehicle speed. The fans can be controlled individually. In order to reduce the load of a single auxiliary converter, at start up the start time delay is different for each fan (3,4,5,6 sec delay)
Depending on the vehicle speed the oil the temperature values to pass from low to high fan speed. The picture belove reports the law to use for fan speed transitions.

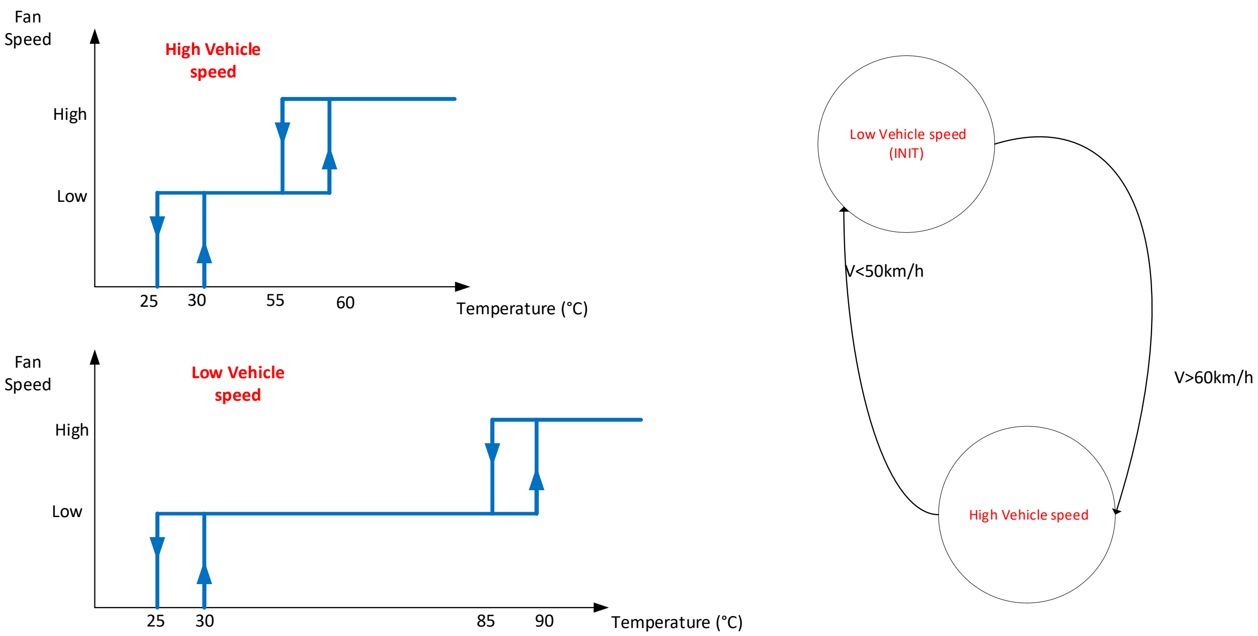


Figure 1 Transformer Fan Cooling Control

3.6.1.1 *Requirements*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.HR34	TCMS monitors the 3-phase bus availability and shall send to Propulsion the enabling to activate the transformer cooling	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR35	The half train Master PCU shall activate the transformer oil pump IF - (AC or DC LCB is closed AND - transformer cooling enable from TCMS AND - Not too low oil level AND - Not overpressure) OR - (Oil temperature is high [AND - transformer cooling enable from TCMS AND - Not too low oil level AND - Not overpressure)	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0

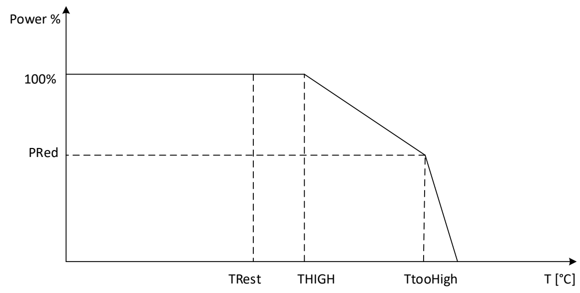
2F_04.01.Zefiro-Europe.CONCEPT.HR36	"Transformer oil pump contactor ON fault or MCB tripped" shall be indicated when the oil pump is activated but the oil pump contactor feedback is not set high	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR37	"Transformer oil pump contactor OFF fault" shall be indicated when the oil pump is not activated but the oil pump contactor feedback is still set high	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR38	IF the oil temperature OR the line inductor estimated temperature is above 30°C AND the fan are switched off THEN the fans shall be activated at low speed	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR39	IF the oil temperature AND the line inductor estimated temperature is belove 25°C THEN the fan shall be switched off	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR40	IF the fan cooling control logic is in the "High vehicle speed" status AND (the oil temperature OR the line inductor estimated temperature is above 60°C) THEN the fans shall be activated at high speed	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR41	IF the fan cooling control logic is in the "High vehicle speed" status AND (the oil temperature OR the line inductor estimated temperature is belove 55°C) THEN the fans shall be activated at low speed	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR42	IF the fan cooling control logic is in the "Low vehicle speed" status AND (the oil temperature OR the line inductor estimated temperature is above 90°C) THEN the fans shall be activated at high speed	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR43	IF the fan cooling control logic is in the "Low vehicle speed" status AND (the oil temperature OR the line inductor estimated temperature is belove 85°C) THEN the fans shall be activated at low speed	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR44	the fan cooling control logic at startup is in the Low Vehicle speed status. When the vehicle speed is above 60km/h the logic moves in High vehicle speed status When When the vehicle speed is belove 50km/h the logic come back to Low vehicle speed status	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0

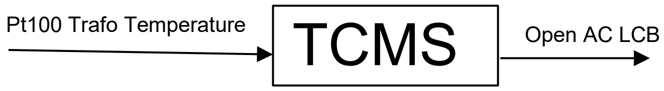
2F_04.01.Zefiro-Europe.CONCEPT.HR45	A minimum period of time [30 sec] has to go between two negative edges of the high speed order. This condition shall have higher priority than the temperature functions. A minimum time to change the speed (10sec) from the last activated	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR46	IF a low fan speed is ordered AND the low speed feedback is not received THEN high speed shall permanently override low speed order	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR47	IF low fan speed is ordered AND the low speed feedback is not received THEN "Transformer fan low speed fault" event shall be created	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR48	IF a high fan speed is ordered AND the high speed feedback is not received THEN low speed shall permanently override high speed order	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR49	IF high fan speed is ordered AND the high speed feedback is not received THEN "Transformer fan High speed fault" event shall be created	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR50	IF oil flow is indicated AND the oil pump is stopped THEN The oil flow sensor is faulty	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR51	IF The oil flow sensor is faulty an envent shall be created	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR52	IF OIL Level is Low an envent shall be created	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR53	IF OIL Level Too Low THEN LCB opening and transformer isolation	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_04.01.Zefiro-Europe.CONCEPT.HR54	IF OIL Level Too Low THEN LCB opening and transformer isolation	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR55	IF OIL Level is Too Low an envent shall be created	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR56	IF a **oil over pressure occurs** THEN LCB opening and transformer isolation	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_04.01.Zefiro-Europe.CONCEPT.HR57	IF a **oil over pressure occurs** THEN LCB opening and transformer isolation	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR58	A IF a **oil over pressure occurs a**n envent shall be created	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)	EVENT	SIL 0

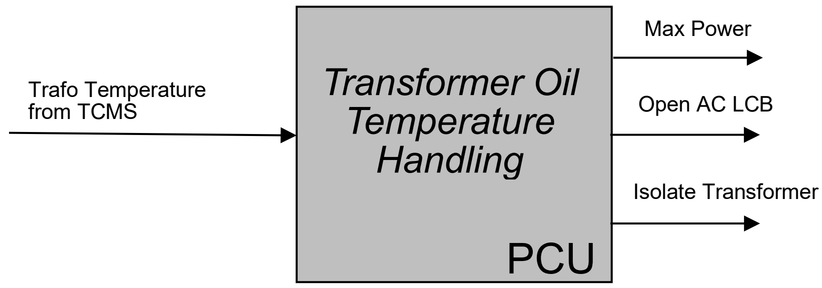
3.6.2 Main Transformer Supervision

The main transformer and line inductors temperatures are monitored by oil temperatures, measured by pt100 sensors acquired from TCMS and sent to PCUs, the primary and secondary transformer temperatures calculated from PCU by an algorithm based on oil temperature and primary and secondary transformer current, the line inductors temperatures calculated from PCU by an algorithm ased on oil temperature and line current
Oil level and oil pressure status are also acquired

	TRest	Thigh	TtooHigh	TExtrHigh
Toil	115°C	130°C	135°C	140°C
Twindings	145°C	160°C	165°C	170°C
Tline inductor	145°C	160°C	165°C	170°C







Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.HR60	The transformer temperature sensors are connected to TCMS safe RIO channels	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_04.01.Zefiro-Europe.CONCEPT.HR61	TCMS shall send the transformer temperature to PCUs.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR62	If both main transformer temperature sensors are faulty or not avalable (e.g. electronic fault) the propulsion effort shall be reset in the half train,(ED and auxiliary converter are available)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

2F_04.01.Zefiro-Europe.CONCEPT.HR324	If both main transformer temperature sensors are fault or not valid the trasnformer fans shall work to maximum speed	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR227	A trasformer temperature sensor shall be considered faulty if the measured Value is out of Range [-50°C 500°C] for 20 seconds	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR228	After 300 second from the transformer cooling enable the temperatures difference mesured by the trasformer pt100s has to be evalueted.It shall be conidered too high if the measured Value differs of more than 20°C for 20 seconds	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_04.01.Zefiro-Europe.CONCEPT.HR229	The reset of the *too High temperature dfference* is possible if the difference becomes lower than 10°C	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR323	If a *too High transformer temperature dfference* is evaluated an event is created	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR63	If one main transformer temperature sensor is faulty or not valid the other sensor shall be used	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.HR327	If one main transformer temperature sensor is faulty or not valid the other sensor shall be used	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR64	If one main transformer temperature sensor is faulty an event shall be created	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR196	Apply a filter on the temperature values to eliminate undue spikes	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR325	If both transformer temperature signals from TCMS are valid and the pt100s not fault the highest value shall be used	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR326	If both transformer temperature signals are valid and the pt100s not fault the highest value shall be used	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 1
2F_04.01.Zefiro-Europe.CONCEPT.HR65	Based on oil temperature and line current the line inductor temperature has to be calculated	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR66	Based on oil temperature, primary and secondary current transformer the primary and secondary windings temperature have to be calculated	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR67	If both main transformer temperature sensors are faulty an event shall be created	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR68	IF the oil temperature is above 130°C OR the winding temperatures are above 160°C OR line inductor temperature is above 160 °C THEN Traction and rigenerative braking shall be reduced linearly from 100% to to 50% when the TtooHigh Temperature value is reached	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR69	IF the oil temperature is above 135°C OR the winding temperatures are above 160°C OR line inductor temperature is above 165 °C THEN Traction and rigenrative braking shall be reduced linearly from 50% to 0% when the TExtrHigh Temperature value is reached	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR70	IF the reduction power due to high temperature is active AND the oil temperature is below 115°C AND the winding temperatures are below 145°C AND line inductor temperature is below	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0

	145 °C THEN the traction and regenerative braking limitation shall be reset				
2F_04.01.Zefiro-Europe.CONCEPT.HR71	IF the oil temperature is above 140°C for 1 second THEN TCMS requires the line trip opening to HVSP. TCMS shall remove the request of line trip opening to HVSP when the oil temperature reaches 135 °C.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 1
2F_04.01.Zefiro-Europe.CONCEPT.HR283	If HVSP receives a line trip opening request from TCMS THEN the line trip shall be open	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL2
2F_04.01.Zefiro-Europe.CONCEPT.HR289	When HVSP has opened the line trip on TCMS request, the opening request can be reset when the reset line trip opening command is received from TCMS	Derived Requirement	4S_04.-.Power Supply (HVSP)		SIL2
2F_04.01.Zefiro-Europe.CONCEPT.HR186	An event is created when HVSP opens the line trip on TCMS request	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	
2F_04.01.Zefiro-Europe.CONCEPT.765	If an oil over temperature (>140°C) in main transformer at AC and DC operation is detected, an EVENT shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	
2F_04.01.Zefiro-Europe.CONCEPT.497	If an oil over temperature (> 130°C) or a winding over temperature (> 160 °C) is detected running AC for 2 times in 12 hours a request to isolate the high voltage section (with the over heated transformer) in AC operation shall be sent to TCMS (refer to section "Isolate high voltage"). 50% of propulsion available	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A
2F_04.01.Zefiro-Europe.CONCEPT.498	If an oil over temperature (> 130 °C) or an over temperature in line inductor (> 160 °C) is detected running DC for 2 times time in 12 hours a request to isolate the high voltage section (with the over heated line inductor) in DC operation shall be sent to TCMS (refer to section 3.4.3 "Isolate high voltage"). 50% of propulsion available.	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		N/A

The following table summarizes the actions requested to protect the transformer

Action	Condition
tractive/braking power reduction	• high transformer oil temperature • high transformer winding temperature • high line inductor temperature • transformer oil cooling fan fault (only low speed available)
tractive/braking power resetting	• Too high transformer oil temperature • Too high transformer winding temperature • Too high line inductor temperature • transformer oil cooling fan fault (low and high speed not available) • no transformer oil flow at low temperature • Transformer oil temperature sensors faulty

LCB open and transformer isolation	• extremely high transformer oil temperature • extremely high transformer winding temperature • Extremely high line inductor temperature • too low oil level • transformer oil flow not normal • transformer overpressure • no transformer oil flow at low temperature during too long time • transformer oil flow sensor fault

3.7 Energy Meter

The energy metering function is a single sub-system function and works independent from any vehicle level function. The vehicle only provides auxiliary power from the DC110V level and generates events if the power supply is lost.
Additionally vehicle can provide the location and time data to the system. The Energy Meter includes a communication unit with capability for direct data transmission of energy stored data to a ground server.

3.7.1 Function Design

Energy Meter System:

- is a combination of the Energy Metering Function (EMF), Data Handling System (DHS), Communication Unit (4G)
- It shall be approved for billing by Eress.
- It is compliant to compliant to TSI 2019 /EN50463:2017It shall be capable of measuring the energy absorbed (consumed) and regenerated by trains.
- The energy data is time-integrated based on a reference period which is selectable and stored on the device internal memory for at least 60 days.
- The vehicle GPS location, date and time data can be correlated with stored energy data.
- Energy Data can be shared with TCMS including real time voltage, current and power readings.
- The diagnostic data is shared with TCMS.
- EMS sends the stored energy data (DHS) to the ground station (Data Collection System DCS) by a dedicated communication physical channel.

3.7.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.796
Description	The vehicle shall provide the location data and time data to the Energy meter system (DHS) for billing purpose.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1313	<u>TCMS: send location and time data to EMS</u> GIVEN the communicaton is established between TCMS and EMS THEN TCMS shall send the location data and time date data to EMS.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.797
Description	EMS sends the energy data (DHS) to the ground station Data Collection System DCS).
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1315	<u>EMS: send energy data to the ground station (DCS)</u> GIVEN	Derived Requirement	4S_04.-.Power Supply (EMS)		SIL 0

	• communication to the ground station (Data Collection System) is established AND • energy data needs to be sent to the ground station (DCS) THEN EMS forward the energy data to ground station (DCS) by its communication unit.				
2F_04.01.Zefiro-Europe.CONCEPT.1319	EMS: send diagnostic data to TCMS GIVEN the EMS is supplied with DC110V supply THEN EMS shall send the diagnostic data to TCMS using TRDP protocol on IP bus	Derived Requirement	4S_04.-.Power Supply (EMS)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR172	EMS: send energy data to TCMS GIVEN the EMS is supplied with DC110V supply THEN EMS forward the energy data to TCMS using TRDP protocol on IP bus	Derived Requirement	4S_04.-.Power Supply (EMS)		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.798
Description	The vehicle shall manage data from/to the Energy meter system (DHS).
Safety level	SIL 0

Nr	**Description**	**Applicability Details**	**Derived to**	**User Interface**	**SIL**
2F_04.01.Zefiro-Europe.CONCEPT.1345	GIVEN TCMS is active THEN TCMS exchanges data with EMS according ICD [1]	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1347	EMS shall be able to receive data from sensors installed in T4 AC HIGH VOLTAGE BOXand in T5 AC HIGH VOLTAGE BOX	Derived Requirement	4S_04.-.Power Supply (EMS)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR32	EMS shall be able to receive data via IP from DC VMT and CMT installed on the roof in T2 and in T7	Derived Requirement	4S_04.-.Power Supply (EMS)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR33	EMS: elaborates energy data of the whole consist independently which (front or rear, AC or DC) and how many pantograph are raised (in DC is possible that 2 pantographes are connected to the line).		4S_04.-.Power Supply (EMS)		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.799
Description	Status of Energy meter system (DHS) and relevant diagnostic events shall be showed on IDU.
Safety level	SIL 0

Nr	**Description**	**Applicability Details**	**Derived to**	**User Interface**	**SIL**
2F_04.01.Zefiro-Europe.CONCEPT.1352	GIVEN TCMS is active WHEN EMS is not in Failure THEN TCMS shall show on IDU the status of EMS <EMS icon> EMS	Derived Requirement	4S_09.03.05.-.TCMS Software	HW HMI	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1353	GIVEN TCMS is active WHEN EMS is in Failure THEN TCMS shall show on IDU the status of EMS <Faulty EMS icon>	Derived Requirement	4S_09.03.05.-.TCMS Software	HW HMI	SIL 0

	EMS				
2F_04.01.Zefiro-Europe.CONCEPT.1355	GIVEN TCMS is active WHEN EMS is in Failure THEN TCMS shall trigger relevant diagnostic events	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

3.8 Key interlocking

The Key Interlocking System is described in the document FF010020369.

3.8.1 Function Design

3.8.1.1 *Normal Condition*

All power supply available: HV, 400V, 110V

3.8.1.2 *Failure Condition*

Failure conditions and failure effects are described in the FMECA-File “FMECA KIS Zefiro Italy_XXX.xls”.

3.8.1.3 *Failure Supervision*

Status of HV-switches and disconnectors is controlled by TCMS and hardwired to control lamp and solenoid protected lock.

3.8.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.679
Description	Two levels of key interlocking shall be supported: – Operation mode – Maintenance mode
Safety level	SIL 2

At Interlocking sequence there are five different modes defined:

- Operation Mode
- Parking Mode
- Disabled Mode
- Depot Supply Mode
- Maintenance Mode

ID	2F_04.01.Zefiro-Europe.TRS.680
Description	In operation mode, high voltage components and auxiliary power supply shall be locked to prevent unwanted and dangerous switching operations.
Safety level	SIL 2

ID	2F_04.01.Zefiro-Europe.TRS.681
Description	Changing from “operation” to “maintenance” mode, the following steps shall be performed: – open line circuit breakers – lowering pantographs – earthing the high voltage equipment, if necessary
Safety level	N/A

1) Operation Mode: HV is live, active MCB is closed, active pantograph is raised, internal 3AC400V is live, 110 VDC is live, depot supply is not possible. KIS is not active.

2a) Parking Mode: is identical with Operation Mode, but the **traction is blocked and the Parking brake is applied**.

2b) Disabled Mode: **HV shut off, all MCBs are open, all pantographs are down, both Isolating/Grounding Switches DC (4a) are in earthed position, depot supply is possible, contactors for 3AC400V are open,** and 110 VDC remains live. Disabled Mode is a parking mode for the train with the possibility to connect depot supply; e.g. for cleaning the train with a vacuum cleaner or charging the 110V batteries. To achieve this mode, the KIS is not active.

3) Depot Supply Mode: HV shut off **and earthed**, all pantographs are down, depot supply is possible, 110 VDC remains live for a limited time; **access to the HV boxes at the roof is possible**.
Depot Supply Mode is a mode, where maintenance work can be performed at cubicles containing live 3AC400V equipment. Also maintenance at roof equipment is possible. To achieve this mode, the KIS is needed.

4) Maintenance Mode: HV shut off and earthed, pantographs are down, **3AC400 V is shut off and earthed, DC-links are earthed**; **access to the key-locked cubicles is possible**, 110 VDC remains live, **depot supply is not possible**.
Maintenance Mode is a mode, where maintenance work can be performed at cubicles containing HV equipment or for exchanging 3AC400V equipment. To achieve this mode, the KIS is needed.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.602	If any A-Key is activated the LCB shall open and the pantograph shall lower	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.682
Description	The hatches of the converter unit must be locked to prevent access to dangerous voltages.
Safety level	SIL 2

3.8.3 Function Earthing of High Voltage Supply

3.8.3.1 *Function Design*

3.8.3.1.1 Normal Condition

The High Voltage system is earthed to secure that maintenance of the system can be done in a secure way without risk for electric shock.

3.8.3.1.2 Failure Condition

Failure conditions and failure effects are described in the FMECA-File “FMECA KIS Zefiro Italy_XXX.xls”.
1) Isolating/Grounding Switch DC (4a) does not close
2) Isolating/Grounding Switch AC (4b) does not close

3.8.3.1.3 Failure Supervision

1) HV DC line is earthed manually by HV DC Earthing Disconnectors 36a and 36b.
2) Status of Isolating/Grounding Switch AC is controlled by PCU (via TCMS) and hardwired to a control lamp and a solenoid protected lock

3.8.3.2 *Requirements*

ID	2F_04.01.Zefiro-Europe.TRS.291
Description	It shall be possible to earth the high voltage supply with a switch remotely controlled. Earthing of the high voltage AC supply shall be done after the line circuit breaker. With this measure the main transformer shall also be earthed.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1016	When the push-button “Lamp Test” from the A1-key Panel in DM1-car is pressed	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

	the Blue lamp “HV AC earthed” at A1-Key-Panel will light.				
2F_04.01.Zefiro-Europe.CONCEPT.1017	When the push-button “Lamp Test” from the A2-key Panel in DM8-car is pressed the Blue lamp “HV AC earthed” at A2-Key-Panel will light.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.147	When the Key from the A-key Panel in one DM-car is moved to the Depot-Supply position the AC- and DC-LCBs shall be ordered to open and all pantographs shall be lowered.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1060	When the Key from the A-key Panel in one DM-car is moved to the Depot-Supply position the AC- and DC-LCBs shall be ordered to open and all pantographs shall be lowered.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1018	When the Key from the A-key Panel in one DM-car is moved to the Depot-Supply position all 3AC400V contactors 18a, 18c and 39 shall open.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1078	When the Key from the A-key Panel in one DM-car is moved to the Depot-Supply position all 3AC400V contactors 18a shall open.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1218	when the A-key is restored to 'normal mode' position (from 'Depot mode') all MV contactors 18c (medium voltage contactor 7) and 39 (medium voltage contactor 6) and also the medium voltage contactor 8 shall be closed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1040	When the Key from the A-key Panel in one DM-car is moved to the Depot-Supply position pre charging devices 29a, 29b, 32a, 32b and disconnectors 33a, 33b shall open.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1079	When the Key from the A-key Panel in one DM-car is moved to the Depot-Supply position pre charging devices 29a, 29b, 32a, 32b and disconnectors 33a, 33b shall open.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1019	When the A1-Key at A1-Key Panel in DM1-car or the A2-Key at A2-Key Panel in DM8-car is moved to the Depot-Supply position the Grounding switches 4a (TT2 and TT7-car) and 4b (T4 and T5-car) shall be ordered to close the earthing contacts.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1080	When the A1-Key at A1-Key Panel in DM1-car or the A2-Key at A2-Key Panel in DM8-car is moved to the Depot-Supply position the Grounding switches 4a (TT2 and TT7-car) and 4b (T4 and T5-car) shall be ordered to close the earthing contacts.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1020	When the A1-Key or the A2-Key is moved to the Depot-Supply position the Blue lamps “HV AC earthed” at both A-Key-Panels in DM1- and DM8-car will light, if Grounding switches AC 4b in T4- and T5-car are in the earthed position.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1021	When the A1-Key or the A2-Key is moved to the Depot-Supply position the solenoid protected locks for the A-Keys at HV DC Earthing Disconnector 36a and 36b (TT2- and TT7-car) will be unlocked, if Grounding switches AC 4b in T4- and T5-car are in the earthed position.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.413	The status of the Grounding switches 4a and 4b shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

2F_04.01.Zefiro-Europe.CONCEPT.414	TCMS sends the A-key status to Propulsion on IP bus	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR174	If A-key is moved to the Depot-Supply the System Isolation Switch shall be ordered for earthing	Derived Requirement	4S_04.-.Power Supply (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.417	The Grounding switch close order signal shall be set low as soon as the feedback Grounding switch 4a and 4b CLOSED goes high.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.419	The Grounding switch open order signal shall be set low as soon as the feedback Grounding switch 4a and 4b OPEN goes high.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.420	The maximum time for close or open Grounding switch order shall be 5seconds.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.421	If the A-key has not been turned to the Depot - Supply position for >7days, then there shall be an event displayed on the IDU. The event shall be reset when the key has been in the Depot - Supply position and the Grounding switches 4a (open when earthed) and 4b (closed when earthed) have followed the right sequence when the A-key is returned to the normal position (4a remains open till pantograph command, 4b is open when not earthed). This ensures that the key is operated to test the Grounding switches function at least every 7 days.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.422	The AC Grounding Switches 4b shall not be able to close when any pantograph is raised.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.755	The AC Grounding Switch 4b shall not be ordered to close when any pantograph is raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.412	When the input from the A-key panel in both DM-cars is not in the Depot- Supply position the AC Grounding switches 4b (T4 and T5-car) shall be ordered to open.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Earth HV DC line manually (left side of the train):
Actions: Manual actions

- Remove A1-key from A1-key panel and put it into the lock of HV DC Earthing Disconnector panel switch 36a (Z-switches in car TT2) and turn it.
- Turn 36a (Z-switches) into earthing position

Results: mechanical interlocking

- The left DC-HV box is earthed by disconnector 36a (Z-switches) manually
- One E1-keys is released for turning and taking off; mechanically
- One B1-key is released for turning and taking off; mechanically

The A1-key cannot be removed if the switch 36a is in earthed position; mechanically

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1024	The position of HV DC Earthing Disconnector 36a (Z-switches) shall be monitored by TCMS with two auxiliary contacts.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

Earth HV DC line manually (right side of the train):
Actions: Manual actions

- Remove A2-key from A2-key panel and put it into the lock of HV DC Earthing Disconnector panel switch 36b (Z-switches in car TT7) and turn it.
- Turn 36b (Z-switches) into earthing position

Results: mechanical interlocking

- The right DC-HV box is earthed by disconnector 36b (Z-switches) manually
- One E2-keys is released for turning and taking off; mechanically
- One B2-key is released for turning and taking off; mechanically

The A2-key cannot be removed if the switch 36b is in earthed position; mechanically

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1026	The position of HV DC Earthing Disconnector 36b (Z-switches) shall be monitored by TCMS with two auxiliary contacts.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

3.8.4 Function Earthing of Converter boxes

3.8.4.1 *Function Design*

3.8.4.1.1 Normal Condition

The High Voltage system is earthed to secure that maintenance of the system can be done in a secure way without risk for electric shock.

3.8.4.1.2 Failure Condition

Failure conditions and failure effects are described in the FMECA-File “FMECA KIS Zefiro Italy_XXX.xls”.

1) The earthing switch does not close on order, that is the system is not earthed

3.8.4.1.3 Failure Supervision

1) Key to converter unit can not be released if the earthing switch is not in earthed position.

3.8.4.2 *Requirements*

ID	2F_04.01.Zefiro-Europe.TRS.683
Description	An earth switch shall be installed near each converter unit. The earth switch is operated to earth the internal connections (DC-link) of the converter unit in maintenance mode. In order to access the converter unit hatches, the earth switch shall be operated, which gives access to a key to the hatch.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.151	An earth switch shall be installed on the DC link of the converter to discharge all capacitors before the maintenance on the converter units.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1054	**Blocking 3AC400V (right side and left side of the train):** Actions: manual action • Turn B1-key at the HV DC Earthing Disconnector panel 36a Results: • Blocking 3AC400V will be initiated by opening 18b (Depot Supply Contactor) **on both sides of the train**. The C1-key can be released. mechanically	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

2F_04.01.Zefiro-Europe.CONCEPT.1055	**Blocking 3AC400V (right side and left side of the train):** Actions: manual action • Turn B2-key at the HV DC Earthing Disconnector panel 36b Results: • Blocking 3AC400V will be initiated by opening 18b (Depot Supply Contactor) **on both sides of the train**. The C2-key can be released. Mechanically	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

Earthing 3AC400V (left side of the train):
Actions: manual action
- take B1-key, put it into the 400V Earthing Disconnector 37a of car T4, and turn it.
- Turn handle of 37a and earth.

Results:
- The B1-key cannot be removed; mechanically
- two C1-keys will be released; mechanically

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1030	The position of 400V Earthing Disconnector 37a shall be monitored by TCMS with two auxiliary contacts.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1081	When the 400V Earthing Disconnector is closed it shall be displayed on IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

Earthing 3AC400V (right side of the train):
Actions: manual action
- take B2-key, put it into the 400V Earthing Disconnector 37b of car T5, and turn it.
- Turn handle of 37b and earth.

Results:
- The B2-key cannot be removed; mechanically
- two C2-keys will be released; mechanically

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1031	The position of 400V Earthing Disconnector 37b shall be monitored by TCMS with two auxiliary contacts.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1057	When the 400V Earthing Disconnector is closed it shall be displayed on IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

Earthing the DC-links (left side, cars DM1 and M3):
Actions: manual action
- Take the two C1-keys, put them into the locks of the two DC-Link Earthing Disconnectors 35a (cars DM1 and M3), and turn them.
- Turn handles of the earthing disconnectors and earth.

Results:
- DC links and HV path (step down chopper (11)) are earthed.
- The C1-keys cannot be released; mechanically
- Two D1-keys from car DM1 and two D2-keys from car M3 are released; mechanically

Earthing the DC-links (right side, cars DM8 and M6):
Actions: manual action
- Take the two C2-keys, put them into the locks of the two DC-Link Earthing Disconnectors 35b (cars DM8 and M6), and turn them.
- Turn handles of the earthing disconnectors and earth.

Results:
- DC links and HV path (step down chopper (11)) are earthed.
- The C2-keys cannot be released; mechanically
- Two D3-keys from car DM8 and two D4-keys from car M6 are released; mechanically

Getting the F-Keys (left side)
Actions: left side, at car DM1; manual action
- Take one of the D1-, D2-, D3- and D4-keys and put all four keys into the F Key Multiplier at car DM1
- Turn them all

Result:
- 12 F-keys are released
- The D1-, D2-, D3- and D4-keys cannot be released, if at least one F-key is taken off.

Getting the F-Keys (right side)
Actions: right side, at car DM8; manual action
- Take one of the D1-, D2-, D3- and D4-keys and put all four keys into the F Key Multiplier at car DM8
- Turn them all

Result:
- 12 F-keys are released
- The D1-, D2-, D3- and D4-keys cannot be released, if at least one F-key is taken off.

Bringing back the train into Operation Mode
The sequence of the KIS has to be performed into the reverse direction for each side of the train.
First all hatches protected by the F-keys have to be closed. The F-keys have to be taken back into its key multipliers in order to get access to the D1-, D2-, D3 and D4-keys. (mechanically)
The D1-, D2-, D3- and D4-keys have to be brought back into the DC-Link Earthing Disconnectors 35a+b, the earthing handles have to be brought into the "not earthed" position and then the C1- and C2-keys are released. (manually)
With the two C1-keys (and the two C2-keys) the 400V Earthing Disconnectors 37a+b for 3AC400V are reset by switching into the "not earthed" position. So, the B1- and B2-keys are released. (manually)
The B1/B2-keys and the E1/E2-keys have to be brought back into the disconnector panels 36a+b and turned, and then switches 36a+b must be turned into the "not earthed" position which will release the A1 and A2-keys. (manually)

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1032	**Earthing the DC-links**: Actions: manual action • Turn C1-key at the HV DC Earthing DC-link panel (DM1) Results: • Turn the lever in earthing position D1-key can be released. Mechanically. The other D keys (D2,D3,D4) can be released repeating the operation on the Dc link earthin panel of the other motor cars Actions:	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.HR192	The position of each HV DC-link Earthing shall be monitored by the associated PCU.	Derived Requirement	4S_05.-.Propulsion		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR193	Each PCU send the earthing dc-link status to TCMS	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.HR194	When the dc-link is earthed it shall be displayed on TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software	Event	

Now we are back in Operation Mode.

The A1/A2-keys must be brought back into the A-key panels 40a/b and turned. This opens the Grounding switches 4b from AC-MCBs in cars T4 and T5.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1036	**To put the converter in Traction Mode:** Actions: take F-keys and put them into its key multipliers in order to get access to the D1-, D2-, D3 and D4-keys. (mechanically) Take D1-, D2-, D3- and D4-and put them into the DC-Link Earthing panel and handle the lever into "not earthed" position Results dc-link in not earthed and the C1- and C2-keys are released. (manually)	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		
2F_04.01.Zefiro-Europe.CONCEPT.195	**To put the 400V bus in not earthed mode** Actions: Put the two C1-keys (and the two C2-keys) in the 400V Earthing panel and handle the lever into "not earthed" position Results: 400V bus in not earthed and the B1- and B2-keys are released. (manually)	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		
2F_04.01.Zefiro-Europe.CONCEPT.1038	After opening of the Grounding switches 4b (left and right AC-MCB) the solenoids in the HV DC Earthing Disconnectors 36a and 36b must switched off and the lock from A1-Key and A2- Key is blocked.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1039	After opening of the Grounding switches 4b (left and right AC-MCB) the Blue lamps “AC HV earthed” at A1- and A2-Key-Panel will switch off.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

Now the Disabled Mode is achieved.
To bring the train into Parking Mode the Tumbler Switches “pantograph up” and “MCB closed” have to operated.
From Parking Mode the train can be brought back into Operation Mode by releasing the parking brakes and releasing the traction control.

3.9 HVAC pre-conditioning on DC line

A Pre-Conditioning (PC) functionality shall be foreseen on DC lines in order to allow train to supply all the needed power to HVAC system without exceeding current limitation (200 Amps per pantograph on 3kVdc, 300 Amps per pantograph on 1.5Vdc networks)). This operation mode in DC needs to raise both the DC pantographs and close both the LCBs.
Pre conditioning functionality will be requested to HVAC under DC line voltage only when requested by the driver. On AC lines instead, PC mode will be automatically managed by TCMS

3.9.1 Requirements

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1192	A dedicated menu on IDU shall be foreseen to enable Pre-Conditioning function	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1193	TCMS shall check that all following condition are true before let the driver enable PC mode: – Train standstill – Parking brake applied – Any DC system is selected (3kV or 1.5kV)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

	– Both pantographs are down (and all LCB open) – NO panto or LCB are set in faulty/cut-out status – NO ACM are set in faulty/cut-out status – NO Towing mode is active – NO Fire alarm active on the train – communication with both HVSP is active – both HVSP are not faulty – Propulsion system ready ("*Status: Preheating possible*" signal is true)				
2F_04.01.Zefiro-Europe.CONCEPT.1194	PCU shall provide to TCMS interface signal to manage pre-conditioning mode – *Status: Preheating possible* – *Command: Activate Preheating*	Derived Requirement	4S_05.-.Propulsion		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1195	If all following condition are <u>not verified</u>, PCU shall set true "*Status: Preheating possible*" signal: – – DC system is isolated – Any DC system disconnector failure is active – Line trip global relay cannot open (all other line trip related failures are covered above	Derived Requirement	4S_05.-.Propulsion		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1196	If Driver has pressed PC soft key on IDU, TCMS shall: – set "*Command: Activate Preheating*" to all leader PCUs – activate traction block – Block panto raising command till both DC roof line disconnector are open	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1197	When "*Command: Activate Preheating*" signal is set, leader PCUs shall open both DC line roof disconnector and configure Line Trip chain in Local mode	Derived Requirement	4S_05.-.Propulsion		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1198	In case one or both panto does not raise on order a warning to the driver shall be displayed that PC is not possible	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1199	In PC mode, with both panto up (and roof disconnector open) when driver commands LCB closure, order shall be set in both half train	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1200	In case one or both LCB does not close on order a warning to the driver shall be displayed that PC is not possible	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1201	When all ACM have started (in operation), TCMS shall activate "*Preconditioning mode request*" in all HVAC units	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1202	If within 60 sec from LCB closure, ALL ACM are not in operation status, LCB shall be opened by TCMS and a warning displayed to the driver (driver is free to choose if re-close LCB and try again or abort PC mode definitively)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1203	If AC mode is selected, pre-conditioning mode has to be managed automatically by TCMS (according HVAC SFRS - 2F_15.02.Zefiro-Europe.TRS.61)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

Abort pre-conditioning mode

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1204	An advice "preconditioning finished" shall be displayed to the driver when all HVAC go out of "*Preconditioning*" status	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1205	If driver presses Pre-conditioning softkey, when PC mode is active, "*Preconditioning mode*" shall	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

	be reset to HVAC ("*Automatic mode*" will be enabled), then request to open LCB and lower panto has to be displayed on IDU				
2F_04.01.Zefiro-Europe.CONCEPT.1206	When both panto are lowered after driver stop PC mode via IDU, TCMS will reset "*Command: Activate Preheating*"	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1207	When "*Command: Activate Preheating*" signal is reset by TCMS, leader PCU shall close both DC line roof disconnector and reset Line trip chain configuration	Derived Requirement	4S_05.-.Propulsion		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1208	In case any precondition is missing (except "*Both pantographs are down*"), TCMS shall – open LCB and reset "*Preconditioning mode request*" signal to all HVAC units – display an warning to the driver that pre-conditioning is not possible – after LCB is open request to lower pantographs shall be displayed to the driver and when driver lowers panto "*Command: Activate Preheating*" shall be set LOW to leader PCUs	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1209	In case during PC mode, one (or both) LCB is opened by local protection (but *Status: Preheating possible* signal is still set), second LCB shall be opened by TCMS and a warning displayed to the driver (driver is free to choose if re-close LCB and try again or abort PC mode definitively)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1210	In case during PC mode, Line Voltage ≤ 2,2 kV DC on 3kV lines (or ≤1,4kV on 1,5kV lines), if it doesn't rise over the threshold within 30s TCMS shall open all LCB (all parameters, time and voltage values, shall be tunable during commissioning) When line voltage comes back and it is over the threshold for more than 30s, LCB shall be automatically closed and pre-conditioning can start again (TCMS shall wait that ALL ACM are in operation and then sent "*Preconditioning mode request*" to all HVAC) (all parameters, time and voltage values, shall be tunable during commissioning) TCMS shall set a warning to the driver describing that pre-conditioning is not possible	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1211	In case during PC mode, 1 ACM is stopped/faulty and it does not restart within 30s, allLCB shall be opened by TCMS and a warning displayed to the driver (driver is free to choose if re-close LCB and try again or abort PC mode definitively)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1212	In case during PC mode, at least 2 ACM are stopped/faulty, all LCB shall be immediately opened by TCMS and a warning displayed to the driver (driver is free to choose if re-close LCB and try again or abort PC mode definitively)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

4 Signal Interface

4.1 I/O Interface

Nr	Signal name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.201	BYP_HVSP	safe	2F_04.01.Zefiro-Europe.CONCEPT.HR177

4.1.1 Supplier system

Inputs:
Outputs:

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.HR80	Panto1 selector: up		2F_04.01.Zefiro-Europe.CONCEPT.980
2F_04.01.Zefiro-Europe.CONCEPT.HR81	Panto2 selector:up		2F_04.01.Zefiro-Europe.CONCEPT.982
2F_04.01.Zefiro-Europe.CONCEPT.HR82	Panto1 selector: down		2F_04.01.Zefiro-Europe.CONCEPT.984
2F_04.01.Zefiro-Europe.CONCEPT.HR83	Panto2 selector:down		2F_04.01.Zefiro-Europe.CONCEPT.983
2F_04.01.Zefiro-Europe.CONCEPT.590	Pantograph disconnector open		2F_04.01.Zefiro-Europe.CONCEPT.566 2F_04.01.Zefiro-Europe.CONCEPT.567 2F_04.01.Zefiro-Europe.CONCEPT.1074
2F_04.01.Zefiro-Europe.CONCEPT.1077	Pantograph disconnector closed		2F_04.01.Zefiro-Europe.CONCEPT.566 2F_04.01.Zefiro-Europe.CONCEPT.567 2F_04.01.Zefiro-Europe.CONCEPT.1074
2F_04.01.Zefiro-Europe.CONCEPT.HR84	main reservoirs pressure switch (PCP)		2F_04.01.Zefiro-Europe.CONCEPT.542 2F_04.01.Zefiro-Europe.CONCEPT.543
2F_04.01.Zefiro-Europe.CONCEPT.604	A-key activated		2F_04.01.Zefiro-Europe.CONCEPT.602
2F_04.01.Zefiro-Europe.CONCEPT.1044	Line voltage selector: AC 25 kV 50 Hz	safe	2F_04.01.Zefiro-Europe.CONCEPT.786 2F_04.01.Zefiro-Europe.CONCEPT.492
2F_04.01.Zefiro-Europe.CONCEPT.1045	Line voltage selector: AC 15 kV 16.7 Hz	safe	2F_04.01.Zefiro-Europe.CONCEPT.492
2F_04.01.Zefiro-Europe.CONCEPT.1046	Line voltage selector: 3 kV DC	safe	2F_04.01.Zefiro-Europe.CONCEPT.492 2F_04.01.Zefiro-Europe.CONCEPT.1329
2F_04.01.Zefiro-Europe.CONCEPT.1047	Line voltage selector: 1.5 kV DC	safe	2F_04.01.Zefiro-Europe.CONCEPT.492
2F_04.01.Zefiro-Europe.CONCEPT.1048	LCB switch: LCB close		2F_04.01.Zefiro-Europe.CONCEPT.979 2F_04.01.Zefiro-Europe.CONCEPT.374
2F_04.01.Zefiro-Europe.CONCEPT.1049	LCB switch: LCB open		2F_04.01.Zefiro-Europe.CONCEPT.979 2F_04.01.Zefiro-Europe.CONCEPT.374

2F_04.01.Zefiro-Europe.CONCEPT.HR85	LCBOpen(da ATP)		2F_04.01.Zefiro-Europe.CONCEPT.1123
2F_04.01.Zefiro-Europe.CONCEPT.HR86	Traction cutoff (da ATP)		2F_04.01.Zefiro-Europe.CONCEPT.1123
2F_04.01.Zefiro-Europe.CONCEPT.HR87	AC ADD bypass request	safe	2F_04.01.Zefiro-Europe.CONCEPT.1291
2F_04.01.Zefiro-Europe.CONCEPT.HR308	DC ADD bypass request	safe	2F_04.01.Zefiro-Europe.CONCEPT.1291
2F_04.01.Zefiro-Europe.CONCEPT.88	PCU (Pantograph Control Unit) major fault		2F_04.01.Zefiro-Europe.CONCEPT.1114
2F_04.01.Zefiro-Europe.CONCEPT.HR309	Emergency operation TL status		2F_04.01.Zefiro-Europe.CONCEPT.797 2F_04.01.Zefiro-Europe.CONCEPT.HR02
2F_04.01.Zefiro-Europe.CONCEPT.HR89	DC Earthing switch		2F_04.01.Zefiro-Europe.CONCEPT.1038

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.544	Pantograph raise order (for each pantograph)		2F_04.01.Zefiro-Europe.CONCEPT.735
2F_04.01.Zefiro-Europe.CONCEPT.1096	Pantograph raise order backup mode (for each pantograph)		2F_04.01.Zefiro-Europe.CONCEPT.1113
2F_04.01.Zefiro-Europe.CONCEPT.597	Fast lowering of all pantographs		2F_04.01.Zefiro-Europe.CONCEPT.296
2F_04.01.Zefiro-Europe.CONCEPT.HR92	LCB Emergency off		2F_04.01.Zefiro-Europe.CONCEPT.1168
2F_04.01.Zefiro-Europe.CONCEPT.HR93	Command: on auxiliary Compressor		2F_04.01.Zefiro-Europe.CONCEPT.542
2F_04.01.Zefiro-Europe.CONCEPT.HR94	AC/DC OverRide (EV8/EV9) (for each pantograph)	safe	2F_04.01.Zefiro-Europe.CONCEPT.1291
2F_04.01.Zefiro-Europe.CONCEPT.HR96	ADD OverRide (PS3) (for each pantograph)	safe	2F_04.01.Zefiro-Europe.CONCEPT.1291
2F_04.01.Zefiro-Europe.CONCEPT.589	Indication lamp: LCB is open		2F_04.01.Zefiro-Europe.CONCEPT.985 2F_04.01.Zefiro-Europe.CONCEPT.986
2F_04.01.Zefiro-Europe.CONCEPT.1075	Command: Close pantograph disconnector		2F_04.01.Zefiro-Europe.CONCEPT.563 2F_04.01.Zefiro-Europe.CONCEPT.1072
2F_04.01.Zefiro-Europe.CONCEPT.1076	Command: Open pantograph disconnector		2F_04.01.Zefiro-Europe.CONCEPT.1073 2F_04.01.Zefiro-Europe.CONCEPT.565
2F_04.01.Zefiro-Europe.CONCEPT.HR97	Command: Close AC Earthing switch		2F_04.01.Zefiro-Europe.CONCEPT.1019 2F_04.01.Zefiro-Europe.CONCEPT.1080
2F_04.01.Zefiro-Europe.CONCEPT.HR98	Command: Open AC Earthing switch		2F_04.01.Zefiro-Europe.CONCEPT.412

4.2 Bus Interface

Supplier to TCMS SW: Propulsion Control -> CCU

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.HR304	Is Half Train Master		2F_04.01.Zefiro-Europe.CONCEPT.HR14
2F_04.01.Zefiro-Europe.CONCEPT.HR315	open line trip active		2F_04.01.Zefiro-Europe.CONCEPT.HR316
2F_04.01.Zefiro-Europe.CONCEPT.HR101	Is_HVMaster		2F_04.01.Zefiro-Europe.CONCEPT.HR15
2F_04.01.Zefiro-Europe.CONCEPT.HR102	DC Voltage detected to AC panto OR AC Voltage detected to DC panto		2F_04.01.Zefiro-Europe.CONCEPT.HR33
2F_04.01.Zefiro-Europe.CONCEPT.HR103	DC Voltage Detected to DC panto		2F_04.01.Zefiro-Europe.CONCEPT.HR26
2F_04.01.Zefiro-Europe.CONCEPT.HR104	AC Voltage detected to AC Panto		2F_04.01.Zefiro-Europe.CONCEPT.HR26
2F_04.01.Zefiro-Europe.CONCEPT.HR300	1500 VDC system selected		2F_04.01.Zefiro-Europe.CONCEPT.HR25
2F_04.01.Zefiro-Europe.CONCEPT.HR301	3000 VDC system selected		2F_04.01.Zefiro-Europe.CONCEPT.HR25
2F_04.01.Zefiro-Europe.CONCEPT.HR302	25 kV AC system selected		2F_04.01.Zefiro-Europe.CONCEPT.HR25
2F_04.01.Zefiro-Europe.CONCEPT.HR303	15 kV AC system selected		2F_04.01.Zefiro-Europe.CONCEPT.HR25
2F_04.01.Zefiro-Europe.CONCEPT.HR105	Status: System Isolation Switch		2F_04.01.Zefiro-Europe.CONCEPT.274
2F_04.01.Zefiro-Europe.CONCEPT.HR107	Status: AC LCB (closed, open ,undefined)		2F_04.01.Zefiro-Europe.CONCEPT.127
2F_04.01.Zefiro-Europe.CONCEPT.HR108	Status: DC LCB (closed, open ,undefined)		2F_04.01.Zefiro-Europe.CONCEPT.127
2F_04.01.Zefiro-Europe.CONCEPT.HR187	Status: DC LCB is faulty		2F_04.01.Zefiro-Europe.CONCEPT.761
2F_04.01.Zefiro-Europe.CONCEPT.HR188	Status: AC LCB is faulty		2F_04.01.Zefiro-Europe.CONCEPT.761
2F_04.01.Zefiro-Europe.CONCEPT.HR109	DCLineCurrent		2F_04.01.Zefiro-Europe.CONCEPT.1130
2F_04.01.Zefiro-Europe.CONCEPT.HR110	ACTransformerCurrentRMS		2F_04.01.Zefiro-Europe.CONCEPT.1130
2F_04.01.Zefiro-Europe.CONCEPT.HR111	DCLineOverCurrent		2F_04.01.Zefiro-Europe.CONCEPT.225
2F_04.01.Zefiro-Europe.CONCEPT.HR112	Status: LocalACLineDisconnector		2F_04.01.Zefiro-Europe.CONCEPT.475

2F_04.01.Zefiro-Europe.CONCEPT.HR113	Status: LocalDCLineDisconnector		2F_04.01.Zefiro-Europe.CONCEPT.475
2F_04.01.Zefiro-Europe.CONCEPT.HR305	DC system disconnector is faulty		2F_04.01.Zefiro-Europe.CONCEPT.1222
2F_04.01.Zefiro-Europe.CONCEPT.HR306	AC system disconnector is faulty		2F_04.01.Zefiro-Europe.CONCEPT.1222
2F_04.01.Zefiro-Europe.CONCEPT.HR318	Isolate AC system		2F_04.01.Zefiro-Europe.CONCEPT.497
2F_04.01.Zefiro-Europe.CONCEPT.HR319	Isolate DC system		2F_04.01.Zefiro-Europe.CONCEPT.498
2F_04.01.Zefiro-Europe.CONCEPT.HR115	Status: MaxACLineCurrent		2F_04.01.Zefiro-Europe.CONCEPT.728
2F_04.01.Zefiro-Europe.CONCEPT.HR116	Status: Max3kVLineCurrent		2F_04.01.Zefiro-Europe.CONCEPT.728
2F_04.01.Zefiro-Europe.CONCEPT.HR117	Status: Max1_5kVLineCurrent		2F_04.01.Zefiro-Europe.CONCEPT.728
2F_04.01.Zefiro-Europe.CONCEPT.HR118	DC line disconnector fault (in opening or status undefined)		2F_04.01.Zefiro-Europe.CONCEPT.997
2F_04.01.Zefiro-Europe.CONCEPT.HR119	AC line disconnector fault (in opening or status undefined)		2F_04.01.Zefiro-Europe.CONCEPT.997
2F_04.01.Zefiro-Europe.CONCEPT.HR120	Status: Ready2close2LCB		
2F_04.01.Zefiro-Europe.CONCEPT.HR121	Status: *Preheating possible*		2F_04.01.Zefiro-Europe.CONCEPT.1193
2F_04.01.Zefiro-Europe.CONCEPT.914	Max traction power due to transformer supervision		
2F_04.01.Zefiro-Europe.CONCEPT.915	Max line current RMS		
2F_04.01.Zefiro-Europe.CONCEPT.916	Command: AC line regeneration inhibit due to transformer supervision		
2F_04.01.Zefiro-Europe.CONCEPT.HR122	Command: Panto down		2F_04.01.Zefiro-Europe.CONCEPT.994
2F_04.01.Zefiro-Europe.CONCEPT.1106	Command: Isolate pantograph		2F_04.01.Zefiro-Europe.CONCEPT.531 2F_04.01.Zefiro-Europe.CONCEPT.761 2F_04.01.Zefiro-Europe.CONCEPT.762 2F_04.01.Zefiro-Europe.CONCEPT.763 2F_04.01.Zefiro-Europe.CONCEPT.995
2F_04.01.Zefiro-Europe.CONCEPT.HR351	Status: AC Earthing switch is faulty		2F_04.01.Zefiro-Europe.CONCEPT.HR350

TCMS SW to Supplier: CCU -> Propulsion Control

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.917	Country Code	No Safe	2F_04.01.Zefiro-Europe.CONCEPT.1013

2F_04.01.Zefiro-Europe.CONCEPT.HR123	Command: 1.5kV DC selected for		2F_04.01.Zefiro-Europe.CONCEPT.HR23 2F_04.01.Zefiro-Europe.CONCEPT.HR12
2F_04.01.Zefiro-Europe.CONCEPT.HR293	Command: 3kV DC selected		2F_04.01.Zefiro-Europe.CONCEPT.HR23 2F_04.01.Zefiro-Europe.CONCEPT.HR12
2F_04.01.Zefiro-Europe.CONCEPT.HR294	Command: 15kV AC selected		2F_04.01.Zefiro-Europe.CONCEPT.HR23
2F_04.01.Zefiro-Europe.CONCEPT.HR295	Command: 25kV AC selected		2F_04.01.Zefiro-Europe.CONCEPT.HR23
2F_04.01.Zefiro-Europe.CONCEPT.HR124	Status: AC local panto up		2F_04.01.Zefiro-Europe.CONCEPT.749
2F_04.01.Zefiro-Europe.CONCEPT.HR296	Status: DC local panto up		2F_04.01.Zefiro-Europe.CONCEPT.749
2F_04.01.Zefiro-Europe.CONCEPT.HR125	Command: MaxACLineCurrent		2F_04.01.Zefiro-Europe.CONCEPT.728 2F_04.01.Zefiro-Europe.CONCEPT.1051
2F_04.01.Zefiro-Europe.CONCEPT.HR126	Command: Max3kVLineCurrent		2F_04.01.Zefiro-Europe.CONCEPT.728 2F_04.01.Zefiro-Europe.CONCEPT.1051
2F_04.01.Zefiro-Europe.CONCEPT.HR127	Command: Max1_5kVLineCurrent		2F_04.01.Zefiro-Europe.CONCEPT.728 2F_04.01.Zefiro-Europe.CONCEPT.1051
2F_04.01.Zefiro-Europe.CONCEPT.918	Status: neutral section		2F_04.01.Zefiro-Europe.CONCEPT.HR265
2F_04.01.Zefiro-Europe.CONCEPT.919	Status: separation section		2F_04.01.Zefiro-Europe.CONCEPT.HR265
2F_04.01.Zefiro-Europe.CONCEPT.934	Status: enable high voltage control from TCMS		2F_04.01.Zefiro-Europe.CONCEPT.HR07 2F_04.01.Zefiro-Europe.CONCEPT.HR12
2F_04.01.Zefiro-Europe.CONCEPT.909	Status:All pantographs down		2F_04.01.Zefiro-Europe.CONCEPT.1227 2F_04.01.Zefiro-Europe.CONCEPT.467 2F_04.01.Zefiro-Europe.CONCEPT.586
2F_04.01.Zefiro-Europe.CONCEPT.HR128	Command: ActivatePreHeating		2F_04.01.Zefiro-Europe.CONCEPT.1196
2F_04.01.Zefiro-Europe.CONCEPT.HR129	Status:A-key Activated		2F_04.01.Zefiro-Europe.CONCEPT.1080
2F_04.01.Zefiro-Europe.CONCEPT.HR130	Enable2LCBClosing		2F_04.01.Zefiro-Europe.CONCEPT.1161
2F_04.01.Zefiro-Europe.CONCEPT.HR131	Open DC roof Disconnector		2F_04.01.Zefiro-Europe.CONCEPT.465 2F_04.01.Zefiro-Europe.CONCEPT.321 2F_04.01.Zefiro-Europe.CONCEPT.1227 2F_04.01.Zefiro-Europe.CONCEPT.255
2F_04.01.Zefiro-Europe.CONCEPT.HR132	Open AC roof Disconnector		2F_04.01.Zefiro-Europe.CONCEPT.465 2F_04.01.Zefiro-Europe.CONCEPT.320 2F_04.01.Zefiro-Europe.CONCEPT.1227 2F_04.01.Zefiro-Europe.CONCEPT.255
2F_04.01.Zefiro-Europe.CONCEPT.HR133	Enable LCB closing		2F_04.01.Zefiro-Europe.CONCEPT.992 2F_04.01.Zefiro-Europe.CONCEPT.HR08
2F_04.01.Zefiro-Europe.CONCEPT.HR307	3-phase voltage bus active to activate main transformer cooling		2F_04.01.Zefiro-Europe.CONCEPT.HR34
2F_04.01.Zefiro-Europe.CONCEPT.HR297	Main Transformer temperqature sensor 1		2F_04.01.Zefiro-Europe.CONCEPT.HR61

2F_04.01.Zefiro-Europe.CONCEPT.HR298	Main Transformer temperqature sensor 2		2F_04.01.Zefiro-Europe.CONCEPT.HR61

Supplier to TCMS SW: HVSP -> CCU

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.HR134	LineVoltageRangeOK	Safe	2F_04.01.Zefiro-Europe.CONCEPT.HR06
2F_04.01.Zefiro-Europe.CONCEPT.HR135	Is_HVMaster	Safe	2F_04.01.Zefiro-Europe.CONCEPT.HR13 2F_04.01.Zefiro-Europe.CONCEPT.HR204
2F_04.01.Zefiro-Europe.CONCEPT.HR136	AC overVoltage level1	Safe	2F_04.01.Zefiro-Europe.CONCEPT.315
2F_04.01.Zefiro-Europe.CONCEPT.HR136	AC overVoltage level2	Safe	2F_04.01.Zefiro-Europe.CONCEPT.315
2F_04.01.Zefiro-Europe.CONCEPT.HR238	AC excessive Line Voltage (UAcLnExHi)	Safe	2F_04.01.Zefiro-Europe.CONCEPT.1082 2F_04.01.Zefiro-Europe.CONCEPT.HR170 2F_04.01.Zefiro-Europe.CONCEPT.HR171
2F_04.01.Zefiro-Europe.CONCEPT.HR137	DC overVoltage level 1	Safe	2F_04.01.Zefiro-Europe.CONCEPT.316 2F_04.01.Zefiro-Europe.CONCEPT.329
2F_04.01.Zefiro-Europe.CONCEPT.HR240	DC overVoltage level 2	Safe	2F_04.01.Zefiro-Europe.CONCEPT.316 2F_04.01.Zefiro-Europe.CONCEPT.329
2F_04.01.Zefiro-Europe.CONCEPT.HR138	AC underVoltage level 1	Safe	2F_04.01.Zefiro-Europe.CONCEPT.HR210
2F_04.01.Zefiro-Europe.CONCEPT.HR239	AC underVoltage level 2	Safe	2F_04.01.Zefiro-Europe.CONCEPT.317
2F_04.01.Zefiro-Europe.CONCEPT.HR139	DC underVoltage	Safe	2F_04.01.Zefiro-Europe.CONCEPT.318 2F_04.01.Zefiro-Europe.CONCEPT.541
2F_04.01.Zefiro-Europe.CONCEPT.HR338	DC underVoltage level 1	Safe	2F_04.01.Zefiro-Europe.CONCEPT.HR336
2F_04.01.Zefiro-Europe.CONCEPT.HR140	AClineVoltageRMS	Safe	2F_04.01.Zefiro-Europe.CONCEPT.321 2F_04.01.Zefiro-Europe.CONCEPT.HR226 2F_04.01.Zefiro-Europe.CONCEPT.HR205
2F_04.01.Zefiro-Europe.CONCEPT.HR141	DCLineVoltage	Safe	2F_04.01.Zefiro-Europe.CONCEPT.321 2F_04.01.Zefiro-Europe.CONCEPT.HR226 2F_04.01.Zefiro-Europe.CONCEPT.HR205
2F_04.01.Zefiro-Europe.CONCEPT.HR142	ACoverCurrent	Safe	2F_04.01.Zefiro-Europe.CONCEPT.225
2F_04.01.Zefiro-Europe.CONCEPT.HR143	ACLineCurrentRMS	Safe	2F_04.01.Zefiro-Europe.CONCEPT.225 2F_04.01.Zefiro-Europe.CONCEPT.1130 2F_04.01.Zefiro-Europe.CONCEPT.HR205
2F_04.01.Zefiro-Europe.CONCEPT.HR144	DC50HzDetected	Safe	2F_04.01.Zefiro-Europe.CONCEPT.112 2F_04.01.Zefiro-Europe.CONCEPT.HR285 2F_04.01.Zefiro-Europe.CONCEPT.HR286
2F_04.01.Zefiro-Europe.CONCEPT.HR145	TrafoDifferentialOverCurrent	Safe	2F_04.01.Zefiro-Europe.CONCEPT.106 2F_04.01.Zefiro-Europe.CONCEPT.HR181 2F_04.01.Zefiro-Europe.CONCEPT.HR276 2F_04.01.Zefiro-Europe.CONCEPT.HR277 2F_04.01.Zefiro-Europe.CONCEPT.758

2F_04.01.Zefiro-Europe.CONCEPT.HR339	S_WngSy	Safe	2F_04.01.Zefiro-Europe.CONCEPT.459
2F_04.01.Zefiro-Europe.CONCEPT.HR146	TrafoCurrentRMS	Safe	2F_04.01.Zefiro-Europe.CONCEPT.106 2F_04.01.Zefiro-Europe.CONCEPT.988
2F_04.01.Zefiro-Europe.CONCEPT.HR147	ReturnTrafoCurrentRMS	Safe	2F_04.01.Zefiro-Europe.CONCEPT.988
2F_04.01.Zefiro-Europe.CONCEPT.HR148	Self-TestRequest (S_TstTimeout)	Safe	2F_04.01.Zefiro-Europe.CONCEPT.989
2F_04.01.Zefiro-Europe.CONCEPT.HR149	Self-testRunning	Safe	2F_04.01.Zefiro-Europe.CONCEPT.1131 2F_04.01.Zefiro-Europe.CONCEPT.1132
2F_04.01.Zefiro-Europe.CONCEPT.HR150	Self-TstFlt	Safe	2F_04.01.Zefiro-Europe.CONCEPT.989 2F_04.01.Zefiro-Europe.CONCEPT.239
2F_04.01.Zefiro-Europe.CONCEPT.HR198	AcFlt	Safe	2F_04.01.Zefiro-Europe.CONCEPT.HR179 2F_04.01.Zefiro-Europe.CONCEPT.239
2F_04.01.Zefiro-Europe.CONCEPT.HR199	DcFlt	Safe	2F_04.01.Zefiro-Europe.CONCEPT.HR180 2F_04.01.Zefiro-Europe.CONCEPT.239
2F_04.01.Zefiro-Europe.CONCEPT.HR203	Flt	Safe	2F_04.01.Zefiro-Europe.CONCEPT.HR202 2F_04.01.Zefiro-Europe.CONCEPT.239
2F_04.01.Zefiro-Europe.CONCEPT.HR151			
2F_04.01.Zefiro-Europe.CONCEPT.910			
2F_04.01.Zefiro-Europe.CONCEPT.911	Status: Line trip		
2F_04.01.Zefiro-Europe.CONCEPT.913			
2F_04.01.Zefiro-Europe.CONCEPT.915	Max line current RMS	Safe	
2F_04.01.Zefiro-Europe.CONCEPT.HR215	LCBClosing not Request AND LCBHoldFeedback High (S_LcbClL&LcbHldFbH)		2F_04.01.Zefiro-Europe.CONCEPT.HR269
2F_04.01.Zefiro-Europe.CONCEPT.HR217	LCBClosingRequest&VLineNotOK (S_LcbCl&VlNok)		2F_04.01.Zefiro-Europe.CONCEPT.HR272
2F_04.01.Zefiro-Europe.CONCEPT.HR290	LCBClosingRequest&WrongCountryCode (S_LcbClH&WngCntCd)		2F_04.01.Zefiro-Europe.CONCEPT.HR271
2F_04.01.Zefiro-Europe.CONCEPT.HR220	LineTripClosed AND LineTrip Feedback Low (S_LineTripCl&FbL)		2F_04.01.Zefiro-Europe.CONCEPT.HR274
2F_04.01.Zefiro-Europe.CONCEPT.HR221	LineTripOpen AND LineTrip Feedback High (S_LineTripCl&FbL)		2F_04.01.Zefiro-Europe.CONCEPT.HR275
2F_04.01.Zefiro-Europe.CONCEPT.916			
2F_04.01.Zefiro-Europe.CONCEPT.1106			

2F_04.01.Zefiro-Europe.CONCEPT.HR291	first HVSP self test warning (S_TstWa1)		2F_04.01.Zefiro-Europe.CONCEPT.HR173
2F_04.01.Zefiro-Europe.CONCEPT.HR292	second HVSP self test warning (S_TstWa2)		2F_04.01.Zefiro-Europe.CONCEPT.HR234
2F_04.01.Zefiro-Europe.CONCEPT.HR313	Supervision Disabling feedback: S_AcLnVCheckDis, S_DcLnVCheckDis, S_ITrafoDiffDis, S_AcLnOCurrDis, S_UVAcLnDis, S_UVDcLnDis, S_AcOvDis, S_DcOvDis, S_IAcLcbDis		2F_04.01.Zefiro-Europe.CONCEPT.HR311

TCMS SW to Supplier: CCU-> HVSP

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.HR152	Country Code	Safe	2F_04.01.Zefiro-Europe.CONCEPT.HR01
2F_04.01.Zefiro-Europe.CONCEPT.HR225	Command: HV Enable		2F_04.01.Zefiro-Europe.CONCEPT.HR18
2F_04.01.Zefiro-Europe.CONCEPT.HR154	Status: local panto up AC		2F_04.01.Zefiro-Europe.CONCEPT.HR04 2F_04.01.Zefiro-Europe.CONCEPT.459
2F_04.01.Zefiro-Europe.CONCEPT.HR299	Status: local panto up DC		2F_04.01.Zefiro-Europe.CONCEPT.HR04 2F_04.01.Zefiro-Europe.CONCEPT.459
2F_04.01.Zefiro-Europe.CONCEPT.HR155	Status: neutral section		2F_04.01.Zefiro-Europe.CONCEPT.HR284
2F_04.01.Zefiro-Europe.CONCEPT.HR156	Status: separation section		2F_04.01.Zefiro-Europe.CONCEPT.HR284
2F_04.01.Zefiro-Europe.CONCEPT.HR157	Command: 1.5kV DC selected	Safe	2F_04.01.Zefiro-Europe.CONCEPT.HR24
2F_04.01.Zefiro-Europe.CONCEPT.HR158	Command: 3kV DC selected	safe	2F_04.01.Zefiro-Europe.CONCEPT.HR24
2F_04.01.Zefiro-Europe.CONCEPT.HR159	Command: 15kV AC selected	safe	2F_04.01.Zefiro-Europe.CONCEPT.HR24
2F_04.01.Zefiro-Europe.CONCEPT.HR160	Command: 25kV AC selected	safe	2F_04.01.Zefiro-Europe.CONCEPT.HR24
2F_04.01.Zefiro-Europe.CONCEPT.HR162	LIM50Hzbypass (Dis50HzSup)	Safe	2F_04.01.Zefiro-Europe.CONCEPT.1151 2F_04.01.Zefiro-Europe.CONCEPT.1148
2F_04.01.Zefiro-Europe.CONCEPT.HR163	LIMbyPass	Safe	2F_04.01.Zefiro-Europe.CONCEPT.HR243 2F_04.01.Zefiro-Europe.CONCEPT.HR175
2F_04.01.Zefiro-Europe.CONCEPT.HR164	Command: HVSPself-test		2F_04.01.Zefiro-Europe.CONCEPT.989
2F_04.01.Zefiro-Europe.CONCEPT.HR165	Enable2LCBClosing		2F_04.01.Zefiro-Europe.CONCEPT.HR29
2F_04.01.Zefiro-Europe.CONCEPT.HR197	HvspTrp	Safe	2F_04.01.Zefiro-Europe.CONCEPT.HR71 2F_04.01.Zefiro-Europe.CONCEPT.HR283
2F_04.01.Zefiro-Europe.CONCEPT.HR197	HvspRes	Safe	2F_04.01.Zefiro-Europe.CONCEPT.HR289

2F_04.01.Zefiro-Europe.CONCEPT.HR312	Supervision Disabling: C_AcLnVCheckDis, C_DcLnVCheckDis, C_ITrafoDiffDis, C_AcLnOCurrDis, C_UVAcLnDis, C_UVDcLnDis, C_AcOvDis, C_DcOvDis, C_IAcLcbDis		2F_04.01.Zefiro-Europe.CONCEPT.HR311

Supplier to TCMS SW: Panto -> CCU

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.942	Pantograph 1 position		2F_04.01.Zefiro-Europe.CONCEPT.284
2F_04.01.Zefiro-Europe.CONCEPT.943	Pantograph 2 position		2F_04.01.Zefiro-Europe.CONCEPT.284
2F_04.01.Zefiro-Europe.CONCEPT.944	Pantograph 1 unintentionally raised		2F_04.01.Zefiro-Europe.CONCEPT.309
2F_04.01.Zefiro-Europe.CONCEPT.945	Pantograph 2 unintentionally raised		2F_04.01.Zefiro-Europe.CONCEPT.309
2F_04.01.Zefiro-Europe.CONCEPT.947	Pantograph 1 unintentionally lowered		2F_04.01.Zefiro-Europe.CONCEPT.309
2F_04.01.Zefiro-Europe.CONCEPT.948	Pantograph 2 unintentionally lowered		2F_04.01.Zefiro-Europe.CONCEPT.309
2F_04.01.Zefiro-Europe.CONCEPT.949	Pantograph 1 does not raise on command		2F_04.01.Zefiro-Europe.CONCEPT.309
2F_04.01.Zefiro-Europe.CONCEPT.950	Pantograph 2 does not raise on command		2F_04.01.Zefiro-Europe.CONCEPT.309
2F_04.01.Zefiro-Europe.CONCEPT.951	Pantograph 1 does not lower on command		2F_04.01.Zefiro-Europe.CONCEPT.309
2F_04.01.Zefiro-Europe.CONCEPT.952	Pantograph 2 does not lower on command		2F_04.01.Zefiro-Europe.CONCEPT.309
2F_04.01.Zefiro-Europe.CONCEPT.953	Pantograph 1 rise time too long		2F_04.01.Zefiro-Europe.CONCEPT.310
2F_04.01.Zefiro-Europe.CONCEPT.954	Pantograph 2 rise time too long		2F_04.01.Zefiro-Europe.CONCEPT.310
2F_04.01.Zefiro-Europe.CONCEPT.955	Pantograph 1 ADD tripped		2F_04.01.Zefiro-Europe.CONCEPT.294 2F_04.01.Zefiro-Europe.CONCEPT.332
2F_04.01.Zefiro-Europe.CONCEPT.956	Pantograph 2 ADD tripped		2F_04.01.Zefiro-Europe.CONCEPT.294 2F_04.01.Zefiro-Europe.CONCEPT.332
2F_04.01.Zefiro-Europe.CONCEPT.957	Pantograph 1 wear detection		2F_04.01.Zefiro-Europe.CONCEPT.1062 2F_04.01.Zefiro-Europe.CONCEPT.1063
2F_04.01.Zefiro-Europe.CONCEPT.958	Pantograph 2 wear detection		2F_04.01.Zefiro-Europe.CONCEPT.1061 2F_04.01.Zefiro-Europe.CONCEPT.1062

TCMS SW to Supplier: CCU -> Panto

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.960	Train speed		2F_04.01.Zefiro-Europe.CONCEPT.1111

2F_04.01.Zefiro-Europe.CONCEPT.HR169	Ice –scraping mode		2F_04.01.Zefiro-Europe.CONCEPT.1111
2F_04.01.Zefiro-Europe.CONCEPT.961	Selected Voltage		2F_04.01.Zefiro-Europe.CONCEPT.1111
2F_04.01.Zefiro-Europe.CONCEPT.962	Selected catenary		2F_04.01.Zefiro-Europe.CONCEPT.1111
2F_04.01.Zefiro-Europe.CONCEPT.963	Pantograph position command (raise order)		2F_04.01.Zefiro-Europe.CONCEPT.283
2F_04.01.Zefiro-Europe.CONCEPT.1120	Wear detection blocked (ICD Weardetec)		2F_04.01.Zefiro-Europe.CONCEPT.1119

Supplier to TCMS SW: ATP -> CCU

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.HR166	CloseLCB		2F_04.01.Zefiro-Europe.CONCEPT.1123
2F_04.01.Zefiro-Europe.CONCEPT.HR167	RaisePantograph		2F_04.01.Zefiro-Europe.CONCEPT.1123
2F_04.01.Zefiro-Europe.CONCEPT.HR168	VoltageChange		2F_04.01.Zefiro-Europe.CONCEPT.196

Supplier to TCMS SW: EMS -> CCU

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.1321	Diagnostics		2F_04.01.Zefiro-Europe.CONCEPT.1319
2F_04.01.Zefiro-Europe.CONCEPT.1322	Energy data (for ground station DCS)		2F_04.01.Zefiro-Europe.CONCEPT.1315

TCMS SW to Supplier: CCU -> EMS

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.1324	location (GPS) data		2F_04.01.Zefiro-Europe.CONCEPT.1313
2F_04.01.Zefiro-Europe.CONCEPT.1325	time/date data		2F_04.01.Zefiro-Europe.CONCEPT.1313

5 References

[1] EC130023132, EC130023133_Data Interface (ICD) Energy Metering System

[2] EC130023723, EC130023724 Data Interface (ICD) HVSP